

IT'S NOT (JUST) ABOUT THE MONEY: PAY AND THE VALUE OF WORKING CONDITIONS IN TEACHING*

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Abstract

This paper quantifies the extent to which teachers earn rents, using a new approach that explicitly takes both pay and non-wage job attributes into account. Using quasi-experimental designs, novel administrative data, and a choice experiment, I estimate the gap in pay and the gap in the value of working conditions between teaching and teachers' next-best jobs, which sum to the teaching rent. Employing variation in who enters and exits teaching induced by cutoff scores in the teacher certification process and shocks to school staffing, I show that teaching pays a premium of around \$20,000 per year (33–40% of the teaching salary) more than teachers' next-best options. Importantly however, I also find that teachers value the working conditions in their next-best jobs more than the conditions in teaching, implying much of the pay premium is a compensating differential. Inexperienced teachers, who report the most difficult teaching conditions, are willing to pay the most for better conditions. My results indicate that teachers earn a large pay premium that is significantly offset by the job's relatively undesirable conditions, resulting in a moderate 16% rent for experienced teachers and no rent for inexperienced teachers. Extending my point estimates to other states suggests that over \$120 billion is being spent annually to compensate teachers for poor working conditions in the US, consistent with reports of teacher shortages today.

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I. INTRODUCTION

Economists have long been interested in measuring worker rents as a means to assess imperfect competition in the labor market and inform regulation (Manning, 2011; Card et al., 2018; Azar, Berry, and Marinescu, 2022; Card, 2022), and recent studies have made progress on estimating rents in the entire labor market (Lamadon, Mogstad, and Setzler, 2022; Jäger et al., 2024). However, while these estimates point to positive rents on average, they mask potential heterogeneity in occupation-specific rents, which may be small or even zero in occupations characterized by undesirable non-wage attributes or labor shortages that may warrant regulatory attention. This paper proposes an approach for assessing an occupation-specific rent by separately evaluating the wage and non-wage components of the rent. I use the approach to assess the extent to which public school teachers earn rents.

Teaching is the largest public sector occupation in the US and teacher salaries account for over half of public school spending, totaling over \$500 billion in 2020–21. Given that state budgets are constrained, the opportunity cost of paying teachers rents is high. However, there are reasons both to believe and to doubt that teachers earn rents. On one hand, public school teaching jobs are associated with job stability and strong unions, which existing studies find may raise teacher compensation without raising student achievement (Hoxby, 1996). On the other hand, widespread reports of “teacher shortages” (Sutcher, Darling-Hammond, and Carver-Thomas, 2016) and evidence that teachers earn less than the average college-educated worker (Allegretto and Mishel, 2018) suggest that teachers may not earn rents at all.¹ Consistent with this uncertainty, existing studies find mixed evidence on how teaching compensation compares to that of teachers’ next-best options.²

However, many of these studies estimate the observational difference in pay between teaching and other jobs, which does not, on its own, identify the teaching rent.³ Furthermore, by focusing on pay differences between teaching and other jobs, these studies overlook a crucial aspect of compensation: the value of working conditions. Given that workers place substantial value on non-wage aspects of jobs (Mas and Pallais, 2017; Maestas et al., 2023), it is necessary

¹For example, see Eliza Fawcett and Jacey Fortin, “How Bad Is the Teacher Shortage? Depends Where You Live.” *The New York Times*, June 21, 2023.

²See, for example: Goldhaber et al. (2022); Champion, Mastri, and Shaw (2011); Scafidi, Sjoquist, and Stinebrickner (2006); Stinebrickner (2002).

³For instance, if teachers are negatively selected on earnings potential, it is possible that they earn less than observationally similar workers (a negative observational pay gap) *and* earn more in teaching than in their next-best options (a positive rent).

to account for both pay and the value of non-wage job attributes when assessing rents. Doing so is especially important when studying occupations with undesirable working conditions, where compensation intended as a *compensating differential* could be misinterpreted as a rent. Recent evidence suggests that teaching may be one such occupation, with over 50% of teacher strikes in the past decade demanding better working conditions (Lyon, Kraft, and Steinberg, 2024) amid increasing reports of stress, burnout, and student behavioral issues on the job.⁴

This paper quantifies the extent to which teachers earn rents using a method that explicitly takes both pay and the value of working conditions into account. The approach entails separately estimating the *gap in pay* and the *gap in the value of working conditions* between teaching and teachers’ next-best jobs, which can be summed to obtain the teaching rent.⁵

Estimating the pay gap and working conditions gap requires addressing two key challenges: unobserved selection into jobs and a lack of data on working conditions across jobs. Workers select into occupations for idiosyncratic reasons, making it difficult to identify workers’ next-best jobs using observational data. Even if workers were randomly assigned to teaching, estimating the pay gap would require panel data on workers’ wages and occupations outside of teaching, which is difficult to come by. There is also no systematic data on the incidence of job attributes across jobs that can be used to assess the working conditions gap between specific occupations.⁶

I address the challenges of selection and lack of data by leveraging institutional features of the occupation in question and by running a targeted, low-cost choice experiment survey. I estimate the pay gap using two quasi-experimental designs and a novel panel dataset covering the near-universe of workers in the state of KY. I estimate the working conditions gap by surveying KY teachers and using choice experiments to elicit their willingness to pay for a wide range of attributes. Taken together, my results provide the most comprehensive estimate of an occupation-specific rent in the U.S. to date.

My analysis proceeds in two main steps. First, I estimate the pay gap using a unique panel dataset from KY. I use two quasi-experimental designs—a fuzzy regression kink (RK) design and an event study design—to leverage variation in who enters and exits teaching. Each design has its own advantage: the RK leverages as-good-as-random variation in who becomes a teacher induced by a cutoff score in KY’s teacher certification process, while the event study

⁴For example, see Sejla Rizvic, “Teachers, Facing Increased Level of Stress, Are Burned Out,” *The New York Times*, March 13, 2023.

⁵Jäger et al. (2024) refer to these as the wage-based and non-wage-based components of the worker rent, following Lamadon, Mogstad, and Setzler (2022).

⁶The most relevant dataset that documents differences in working conditions across U.S. workers, the American Working Conditions Survey (AWCS), contains few teachers (Maestas et al., 2023).

leverages the timing of teachers’ separations to other jobs. The designs estimate the pay gap for inexperienced and more experienced teachers, respectively.

My main finding from the quasi-experimental analyses is that becoming a teacher has a large, positive treatment effect on earnings—or in other words, that the teaching job offers a large pay premium compared to teachers’ next-best jobs. The RK shows that for individuals who score near the teaching entry exam cutoff on their first attempt, their next-best in-state options four years later pay an estimated \$18,000–21,000 less per year than teaching does, and fall in one of four industries: education, health care, retail, or accommodation and food services. Similarly, event studies around teacher exits show that exiting teachers earn an average of \$20,000–25,000 less per year four years after exit and tend to leave to the same four industries. The estimates imply a large pay gap of 33%–40% of a KY teaching salary.

Second, I estimate the working conditions gap by fielding a survey to KY teachers to learn about and estimate teachers’ valuations of the working conditions in teaching. To ensure that my estimates can be benchmarked to prior work, I replicate the survey instrument in the American Working Conditions Survey (AWCS) ([Maestas et al., 2023](#)): first asking workers which attributes their jobs offer, and then posing choice experiments in which workers are shown pairs of jobs with randomized attributes and asked which jobs they prefer. With the survey data, I estimate the working conditions gap in three steps. First, I identify the bundles of attributes offered by teaching and teachers’ “next-best” jobs—that is, lower-paying jobs in the education and healthcare sectors in the AWCS, as identified in the RK and event study analysis. Second, I use the choice experiments to estimate teachers’ willingness to pay for each attribute, following the standard methodology in the willingness-to-pay literature ([Wiswall and Zafar, 2018](#)). Finally, I aggregate the estimates to obtain teachers’ total willingness to pay to give up the attributes in teaching and switch to the attributes in their next-best jobs.

My main finding from the survey is that for teachers, teaching offers *less* desirable conditions than teachers’ next-best jobs. While teaching does offer unique attributes that teachers value, such as working with children and “summers off,” the job also does not offer important attributes that teachers value and that are common elsewhere, such as the absence of hostility and a relaxed pace of work. As a result, I find that experienced teachers in KY are willing to pay 17% of their salary to switch to their next-best job solely for more desirable working conditions, while inexperienced teachers are willing to pay much more—51% of their salary—to make the same switch. The difference in magnitudes is driven by inexperienced teachers facing less desirable job conditions than experienced teachers: in particular, inexperienced teachers

do not have high job security (“tenure”), and they face more hostility and less administrative support than experienced teachers do in conflicts with parents and children.

Overall, my findings suggest that teachers earn a large pay premium that is significantly offset by the job’s relatively undesirable working conditions, resulting in experienced teachers earning rents but inexperienced teachers earning no rents at all. My estimates suggest that experienced teachers earn a rent of 16% of their salary—surprisingly similar in magnitude to recent estimates of worker rents in the entire labor market (Lamadon, Mogstad, and Setzler, 2022; Jäger et al., 2024). However, for inexperienced teachers, my results suggest that their pay premium functions entirely as a compensating differential, consistent with the fact that younger teachers experience the highest turnover rates. Extending my results from KY to other states suggests that over \$120 billion is being spent annually to compensate teachers for relatively poor working conditions. As such, identifying cost-effective ways to close the gap in working conditions between teaching and other jobs may be a more pressing policy issue than whether teachers earn rents. Education policy has taken steps that align with this conclusion, with the Biden administration calling on Congress in 2021 to invest \$9 billion in improving teaching conditions in addition to pay.

This paper primarily contributes to two strands of literature. The first strand investigates the extent to which workers earn rents. Most work in this area focuses on quantifying worker rents in the entire labor market (Manning, 2011; Lamadon, Mogstad, and Setzler, 2022; Jäger et al., 2024). However, few papers focus on measuring worker rents in specific industries or occupations, which may differ from the average and be more tangible for wage regulation, especially in public sector jobs.⁷ This paper proposes an approach for quantifying an occupation-specific rent that can be implemented by leveraging institutional features (e.g., cutoffs on certification tests) to identify next-best jobs and using low-cost choice experiments to estimate workers’ preferences for bundles of job attributes.⁸ I apply the method to estimate the teaching rent, providing the most comprehensive estimate of an occupation-specific rent to date.

The second strand studies teacher labor markets. A significant body of work examines how teaching compensation and preferences over teaching-specific job attributes explain teacher sorting across schools (Boyd et al., 2005; Biasi, Fu, and Stromme, 2025; Bates et al., 2025; Johnston,

⁷A notable exception is Kroft et al. (2025), who focus on the construction industry and use features of the industry to estimate an equilibrium model of the labor and product market to evaluate rents. A large literature has sought to understand why wage premia differ across industries (e.g., Card, Rothstein, and Yi (2024)).

⁸A growing body of work uses, and demonstrates the validity of using, choice experiments to understand workers’ preferences for various job attributes, from dignity at work (Dube, Naidu, and Reich, 2025) to workplace flexibility (Drake, Thakral, and Tô, 2022).

2025; Wiswall and Zafar, 2018).⁹ Yet relatively few studies focus on documenting what teachers’ *outside option* compensation and preferences over *outside option* working conditions are, both of which are key factors in workers’ decision to select into teaching and are particularly important to understand given concerns about teacher shortages.¹⁰ One reason for this dearth of evidence is that it is challenging to identify workers’ next-best options in general. This paper provides the first quasi-experimental evidence characterizing teachers’ next-best jobs in terms of pay, working conditions, and industry/occupation. The results suggest that relatively undesirable working conditions are a key reason why some who consider teaching may instead choose another occupation.

My study is most closely related to Maestas et al. (2023), Johnston (2025), and Jäger et al. (2024). Maestas et al. (2023) embed a choice experiment in the AWCS to assess how job attributes and preferences for working conditions vary across U.S. workers. I survey KY teachers using two adapted versions of the AWCS, allowing me to benchmark my findings for teachers to those of Maestas et al. (2023), and crucially, to compare teaching to teachers’ next-best jobs and compare teachers’ preferences to those of other workers. Johnston (2025) is the first to use a choice experiment to assess how teachers value teaching compensation and a wide range of job attributes, including key general attributes such as managerial (i.e., principal) support. I extend Johnston’s approach to examine how much teachers value the bundle of attributes not just in teaching, but also in their next-best jobs, allowing me to estimate the working conditions gap. My estimates of a large compensating differential in teaching corroborate Johnston’s conclusion that “shifting resources into compensation and amenities that teachers prefer relative to cost” may be an effective way to “improve the appeal of teaching.” Finally, Jäger et al. (2024) use a survey to elicit individual workers’ rents and the pay gap between their job and their next-best option in the German labor market. These estimates allow them to assess the accuracy of workers’ beliefs about their outside options. I employ a related approach that uses the same rents definition and use a survey to back out a component of the rent. That said, my approach differs in two key ways. First, I use the survey to back out the non-wage component of the rent, rather than the wage component. Second, I identify workers’ next-best options using not only job switches, but also quasi-experimental variation in who becomes a teacher, made possible

⁹For instance, a number of papers estimate teacher preferences over student characteristics, using teacher assignment data (Boyd et al., 2005; Biasi, Fu, and Stromme, 2025), application data (Bates et al., 2025), or choice experiments (Johnston, 2025; Wiswall and Zafar, 2018).

¹⁰The most closely related work examines how *changing* outside options affect selection into teaching. See, for example, Nagler, Piopiunik, and West (2020), who leverages variation in outside options caused by business cycles, and Bacolod (2007), who uses women’s changing labor market opportunities over the twentieth century.

by leveraging institutional features of the occupational licensing process.

The rest of the paper proceeds as follows. Section II describes how I estimate the pay gap, from data and methods to results. Similarly, Section III describes how I estimate the gap in the value of working conditions. Section IV discusses the implications of my estimates, as well as how the estimates can be reconciled with previous findings. Section V concludes.

II. ESTIMATING THE PAY GAP

II.A. Identification Challenge

The main identification challenge in pinning down the pay gap between teaching and teachers’ next-best job options is selection into teaching: the decision to become a teacher or earn teaching qualifications is endogenous to earnings potential.

I circumvent this challenge in two ways. First, I exploit exogenous variation in whether one gets certified to teach, induced by the cutoff on one of the exams required for entry into KY’s teacher preparation programs. This approach identifies the treatment effect of becoming a teacher or getting certified on short-term earnings, or equivalently, the pay gap between teachers’ next-best options and teaching for inexperienced teachers. Second, I leverage job switches induced by exits from teaching to other jobs. This approach allows me to estimate the initial gap in pay that arises after a teacher moves—in other words, the pay gap between teachers’ next-best options and teaching for experienced, exiting teachers.

The following sections describe the features of the teacher certification process most relevant for identification and the data I use before detailing the empirical strategies and results.

II.B. Key Features of the Teacher Certification Process

In most U.S. states, including KY, the traditional path to becoming eligible to teach in public schools is a multi-year process colloquially known as the “teacher pipeline.”¹¹ The process typically consists of three stages: (1) enroll in an educator preparation program (EPP), which may have an entry exam requirement, (2) graduate from the EPP and earn a state-approved teaching certificate, which may have an exit exam requirement, and (3) apply for, be offered, and accept a position in a public school as a teacher. EPPs are commonly comprised of up to

¹¹The majority of public school teachers today earned their teaching certificates through the traditional pipeline. However, over the past decade, political pressures and concerns of labor shortages in teaching have led to increased demand and supply of “alternative route certifications,” which allow individuals to bypass the traditional pipeline to become “emergency certified.”

two years of coursework, which college students may combine with the last two years of their four-year college degree.¹²

Rather than design their own entry or exit exams, most EPPs use standardized tests designed by the testing institution ETS.¹³ These exams, collectively referred to as the “Praxis” exams, test skills ranging from core understanding of math, reading, and writing, to knowledge of specific fields of study, such as American history or geography. Importantly, although the exams are meant for prospective teachers, not all of the Praxis exams test pedagogical knowledge. Some, like the Praxis CASE (“Core Academic Skills for Educators”) math exam, test middle-school level proficiency in mathematics.

Unlike other exams offered by ETS that are used to apply to a wide range of programs, such as the GRE or the SAT, the Praxis is only ever used to test individuals for entry into or exit from teaching preparation programs. However, states vary in which Praxis exams they require and in what constitutes a passing score. State departments of education are transparent about these differences.¹⁴

I focus on one particular set of Praxis exams: the Praxis CASE, which are the three core exams in math, reading, and writing. The Praxis CASE is a common entry requirement for EPPs. For example, in KY it is mandated that individuals must pass the Praxis CASE to be admitted to an EPP. Most test-takers who attempt one of the three exams choose to sit all three tests at once, a strategy that is also incentivized by ETS’ pricing scheme: test-takers can take one exam for \$90 or all three for \$150. Retaking is allowed and technically unlimited, although test-takers must wait 28 days between attempts and face the same pricing schedule. Based on national Praxis statistics and the pricing schemes of test preparation programs, the math component is the most frequently retaken component.^{15, 16}

The way the test is run makes it difficult for test-takers to target a specific score. Like other ETS exams, the Praxis CASE is multiple-choice, timed, and administered on computers in monitored exam rooms with no Internet access. Each exam question carries its own weight

¹²Some states also require that students engage in a form of teacher training after the EPP before they receive their full certificate, but this is relatively rare.

¹³As of now, twenty-six states require certain passing scores on the Praxis at some point in the teacher pipeline. Others, e.g. Texas, also require exams but write and administer their own state exam.

¹⁴See state department of education and ETS websites. As of April 2023, twenty states/provinces use the Praxis CASE exam in their teacher pipeline, the majority of which utilize a cutoff of 150 on the math exam, 156 on the reading exam, and 162 on the writing component.

¹⁵Magoosh, for example, offers two programs: a 6-month preparation program for all three exams at \$99, and a 6-month math-only program for \$79.

¹⁶Some programs accept students on the condition that they pass the CASE during the program. However, even if conditional admittance is allowed, students must pass all three tests in order to graduate.

corresponding to its difficulty level, which is not revealed to the test-taker, but instead is used to convert the raw scores into standardized scores which range from 100 to 200. Test-takers thus see their scores immediately at the end of the exam. Time pressure is salient as well: test preparation programs advise test-takers on strategies to best manage their time and try to maximize their score rather than aim to pass.

The use of the Praxis CASE in KY has varied over time. Since the mid-2000s, the Educator Professional Standard Board (EPSB), which governs the certification rules for teachers in KY, has required that applicants pass the Praxis CASE to enter state EPPs. During this time period, cutoff rules became standardized throughout KY.¹⁷ Between 2013 and 2023, there were no changes in ETS’ design of the Praxis CASE nor in the common cutoff scores used in KY.

II.C. Matched Employer-Employee Data

To estimate the pay gap, I construct a novel administrative panel dataset covering the near-universe of workers in KY. The panel allows me to track individuals who take the Praxis through to their future employment and to observe worker flows in and out of public schools.

The main variables of interest are individual earnings, industry, Praxis scores, and employment as a teacher. Earnings and industry come from the KY Unemployment Insurance (UI) database, which contains quarterly earnings and 6-digit industry NAICS codes associated with each source of income, for workers between the ages of 18 to 60.¹⁸ Individual Praxis scores come from the Educator Professional Standards Board (EPSB): for each person, these data include the date and scores of all Praxis attempts and annual indicators of the type (e.g. full, emergency, temporary) and validity of their teacher certification status. Indicators for employment in the public school system come from the KY Longitudinal Data System, a database maintained by KYSTATS that contains person-year records on all staff in the public school system provided by the KY Department of Education (KDE) and MUNIS. These data include information on each individuals’ district and school of employment, occupation (e.g. teacher or principal), and links to classrooms and student outcomes, allowing me to observe when in-

¹⁷Prior to 2012, most EPPs in KY required the earlier version of the Praxis CASE, the “Praxis I,” for entry, but set their own cutoff scores. In 2012, the state mandated a common set of cutoff scores for all EPPs. Finally, in 2014, the Praxis I was discontinued by ETS, resulting in the Praxis CASE becoming the only set of required exams for entering programs with common cutoffs across the state. Test-takers could not “mix and match” test scores from the Praxis I and Praxis CASE—to be eligible for an EPP, they had to take the complete suite of Praxis CASE exams.

¹⁸I create a crosswalk between firm names under the education NAICS code and the official list of private schools on the KY Department of Education (KDE) website to identify private schools where possible.

dividuals move between schools or districts or change occupations within public schools. I also construct indicators for postsecondary enrollment, completion, or Pell grant receipt in-state by merging in data from the Council on Postsecondary Education (CPE), and indicators for whether an individual is listed as a parent on a birth certificate for a newborn child from the Vital Statistics records.

The final sample includes the union of all individuals who ever attempted the Praxis CASE in KY between 2008 and 2022 and all individuals who ever appear as employed in the KY UI data system between 2009 until 2022. Demographic information such as gender, race, and ethnicity exists for nearly all individuals.

There are two notable limitations of the UI data that persist in the final data for analysis: the lack of information on hours/out-of-state employment, and on occupations outside of the public school system. As is true of most state UI data systems, the KY UI data cannot be used to separately identify whether someone is unemployed or has left the state. I refer to individuals in either category as “non-employed.” To learn about teachers’ next-best occupations, I supplement the administrative data with the matched sample in the Current Population Survey (CPS), which contains a nationally representative, two-period panel, with observations spaced one year apart. From 2009 to 2023, nearly 5,000 individuals in the matched CPS report being a teacher in the first round but holding a different occupation in the second round.

Table I shows sample means describing all individuals who attempted the Praxis CASE for the first time between 2013 and 2017 in KY.¹⁹ Panel A shows key demographic characteristics. First-time test-takers tend to be in their third year of college and are mostly white women, reflecting the demographic composition of teachers in the state.²⁰ One-fifth hold a Pell grant, and two-thirds are working in the year before the exam.

Panel B summarizes statistics related to Praxis performance and retaking behavior. The majority (80%) of test-takers eventually pass the entire Praxis CASE. However, these passes are not all obtained on the first attempt: for the least-passed component, math, just under two-thirds of test takers pass on the first attempt. The gap between the initial and eventual passing rate is explained by retakes: 80% who do not pass on the first try attempt it again. Only a third pass on their first retake. The majority of first-time attempters retake the exam

¹⁹I focus on individuals who took the exam between 2013 and 2017 because the Praxis CASE was last revised then, and because the full panel ends in 2022, meaning I can only observe employment outcomes four years out for individuals who took the exam before 2018.

²⁰Although the mean age is 24, the median age is 21, consistent with our understanding that individuals are most likely to take the Praxis CASE in the third year of college.

no more than three times.

Finally, Panel C shows that there is significant “leakage” of individuals from the pipeline over time. Only 53% of first-time attempters become certified to teach within four years of taking the Praxis. Most of the leakage occurs at the Praxis stage and at the certification completion stage, while 87% of individuals who get certified work as a teacher shortly thereafter.

II.D. Empirical Strategy I: Regression Kink Design

My main empirical strategy for estimating the pay gap is a fuzzy RK design. The RK leverages variation in who becomes a teacher, generated by the cutoff on the Praxis exam, to identify the treatment effect of becoming a teacher on earnings. The key insight is that, in this environment, the treatment effect is by definition the difference in potential earnings between teaching and their next-best job for interested teachers—or, in other words, the pay gap of interest.

Why an RK and not an RD? Figure I shows the first-stage relationship between first-attempt test scores and whether test-takers eventually pass, earn teacher certification, or become teachers. Visually, there is a strong kink and no discontinuity in the likelihood of becoming certified or a teacher, motivating the use of a fuzzy RK design.

However, it is not obvious *ex ante* that the first-stage would lack a discontinuity: more often than not, quasi-random assignment around an exam score cutoff creates a strong discontinuity in the first-stage, making an RD the appropriate design choice. I therefore begin this section by walking through the three-step reasoning behind why an RK, not an RD, is ultimately the appropriate design in this environment.

Step 1: Conditions for a Sharp RD. Imagine a world in which the Praxis is a one-shot exam and test-takers vary only in their ability to pass the exam. Anyone who fails on their first attempt cannot eventually become a teacher. Provided that individuals cannot precisely manipulate their scores, there is as-good-as-random variation in whether test-takers pass or fail near the cutoff. Figure II(a) shows a simulation of the first-stage relationship between first-attempt Praxis scores and the likelihood of becoming a teacher that would arise in such a world.²¹ In this environment, the sharp RD estimand identifies the treatment effect of becoming a teacher on earnings (Hahn, Todd, and Van der Klaauw, 2001).

Step 2: Conditions for a Fuzzy RD. However, the Praxis is not a one-shot exam: retaking

²¹Simulation details can be found in Supplementary Materials I.B.

is allowed and commonplace. Instead, then, imagine augmenting the above world to allow those who fail initially to retake the exam with equal probability. Because scores are a noisy signal of underlying ability, those who nearly pass on their first attempt are very likely to pass on their retake, those who score further below the cutoff are less likely to pass on their retake, and so on. The resulting first-stage relationship resembles Figure II(b): a small discontinuity may still exist at the cutoff, and a strong kink arises driven by the availability of the option to retake.

The pattern in Figure II(b) is an example of “imperfect compliance” in an RD setting: some fraction of those who fail on their first attempt still become teachers. However, as long as there is randomization in assignment around the cutoff and a pronounced discontinuity, the fuzzy RD estimand identifies the treatment effect of interest. Existing work uses precisely this approach to evaluate the effects of passing various exams on future labor market outcomes.²²

Step 3: Conditions for a Fuzzy RK. There is, however, one additional wrinkle in my setting that deviates from the typical fuzzy RD setting: I find a strong kink but virtually *no discontinuity* in the likelihood of certification at the cutoff, as shown in Figure I(c).

Why is there no discontinuity at the cutoff? If test-takers who initially fail the exam are all equally likely to retake, then a gap should persist, as illustrated in the simulations in Figure II. The lack of gap thus suggests that retaking probabilities are not uniform across test-takers who initially fail. This is indeed what I find: Figure III shows that among test-takers who initially fail, those who score closer to the cutoff are more likely to retake the exam than those who score further away. The slope in the retaking probability to the left of the cutoff is non-trivial: those who score a 146 or 144 are around 5 p.p. less likely to retake the exam than those who score a 148, despite the scores differing by only one or two correct multiple-choice answers. In contrast, virtually none of those who pass on their first-attempt retake the exam.

There are two possible explanations for why the probability of retaking is increasing in first-attempt scores to the left of the cutoff. One explanation is that receiving a higher score, even by a difference of just one multiple-choice question, actually induces test-takers to become more likely to retake the test. This mechanism can be thought of as an “encouragement” effect: all else equal, receiving a higher score signals to the test-taker that they have a higher chance of passing, which “encourages” them to retake the test. The other explanation is selection: people who score a 148 are fundamentally different from those who score a 146, and so on, in ways that explain their different retaking behavior. While it is difficult to test the encouragement

²²For example, [Jepsen, Mueser, and Troske \(2016\)](#) use the first score on the GED as the running variable in a fuzzy RD to evaluate the long term effect of earning a GED on earnings.

effect directly, it is feasible to test for selection. Consistent with quasi-random assignment of test-takers within small bandwidths of test scores, I find that the strong positive relationship between retaking and first-attempt scores persists even after controlling for a host of baseline test-taker characteristics including gender, race, previous earnings, age, and whether one holds a Pell grant. I therefore take the encouragement effect to be the most plausible mechanism underlying the positive slope of the retaking probability to the left of the cutoff.

The lack of discontinuity rules out the use of a fuzzy RD design. However, in cases where there is a kink and no discontinuity, a fuzzy RK design can be used to identify the treatment effect of interest when certain conditions hold (Dong, 2018).²³

Identification. Four assumptions must hold for the fuzzy RK design to identify the treatment effect of becoming a teacher on earnings.

Assumption 1 is that test-takers cannot precisely manipulate their test scores. Under this assumption, there is as-good-as-random assignment of test-takers around the cutoff based on their first-attempt scores. These marginal scorers are comparable in every way except that (1) those who pass can continue directly on to certification, while (2) those who fail must incur additional costs to earn certification, such as extra fees and a waiting period.

Assumption 2 is that there is virtually no discontinuity in the first-stage relationship between first-attempt test scores and becoming a teacher. As shown in Dong (2018), if there is a kink but no discontinuity, a fuzzy RK (but not a fuzzy RD) will identify a weighted average treatment effect.

Finally, Assumptions 3 and 4 are the classic relevance and exclusion restriction assumptions needed in instrumental variables designs. Assumption 3 is that scoring above the cutoff versus below the cutoff dictates a clear change in the slope of the likelihood of treatment. Assumption 4 is that the mechanism causing the change in slope at the cutoff must only affect the outcome of interest (earnings) through its effect on treatment (becoming a teacher).

If these assumptions hold, then within some small bandwidth of scores near the cutoff, the cost of certification is exogenously varied by *whether* one passes and *how far* they score from the cutoff on their first attempt. As such, the interaction between these two measures can be used as a valid instrument for becoming a teacher. This is precisely what the fuzzy RK design does: it leverages the fact that the change in the slope of the first-stage at the cutoff is induced

²³Following Dong (2018), I also show that if there is both a discontinuity and a kink, a fuzzy RD (but not a fuzzy RK) will identify a weighted average treatment effect. See Supplementary Materials I.A for formal proofs.

by exogenous variation in who falls to the right or left (which creates the break around which the slope can differ) and how far they are from the cutoff (which allows for different slopes to the right and left of the cutoff).

I take the lack of discontinuity shown in the first-stage relationships in Figure I as support for Assumption 2. Below, I provide evidence in support of Assumptions 1, 3, and 4.

Evaluating the quasi-experiment. I first present evidence in support of Assumption 1: that is, that the first-attempt scores as good as randomly assigned near the passing threshold.

Smoothness in the running variable. A key assumption for valid inference in the RK design is that the density of the running variable—the first-attempt math score—is smooth through the cutoff. A kink or a jump in the density of the sample at the cutoff would suggest endogenous sorting of individuals around the passing cutoff, invalidating the design. Figure IV(a,b) shows the distribution of first-attempt math scores around the cutoff of 150 for the main analysis sample, along with fitted polynomials to the right and left of the cutoff.²⁴ There distribution does not appear to exhibit a jump or kink at the cutoff.²⁵

Balance. Another implication of no sorting around the threshold is that individuals on either side of the cutoff are “balanced” on pre-determined characteristics prior to taking the Praxis for the first time. I check this implication in three ways: (1) by visually examining the distributions of each of a large set of baseline characteristics over first-attempt scores, including gender, race (Black, White, Hispanic, and others), age, whether one holds a Pell grant, whether one is enrolled in a Master’s or Bachelor’s program, whether one is already working, the industry one is employed in if so (education, retail, healthcare, and others), and the year and month of the exam; (2) by combining the baseline characteristics into a single “covariates index”—i.e., predicting the outcome of interest (earnings, employment, or industry) using baseline characteristics only, as in Card et al. (2015)—and testing whether the index is predictive of the treatment (Figure IV(c,d)), and (3) by conducting paired *t*-tests comparing differences in baseline characteristics of individuals who score within 10–15 points of the cutoff. All three tests suggest that there is balance on baseline characteristics.

²⁴Each bin represents a single score; gaps arise between the bars because only even number scores are possible on all components of the Praxis CASE.

²⁵Because the test scores can only take on integer values, the formal McCrary test (McCrary, 2008) for manipulation around the cutoff reports discontinuities at every point.

Evaluating instrument validity. I now provide evidence in support of Assumptions 3 and 4, which combined suggest that the proposed IV—the interaction of whether one passes on their first attempt and how far they score from the cutoff on their first attempt—is valid.

Assumption 3 (relevance) can be assessed by visually examining the first-stage patterns in Figure I, which clearly show that the slope of the likelihood of becoming a teacher sharply decreases at the cutoff. More formally, estimating the first-stage equation defined below shows the decline in the slope is significant and negative, suggesting that the IV is relevant.

For Assumption 4 (the exclusion restriction) to be satisfied, it must be the case that the mechanism causing the change in slope at the cutoff only affects earnings through its effect on the likelihood of becoming a teacher. As outlined earlier, the change in slope at the cutoff arises from three features of the Praxis-taking process: (1) being quasi-randomly assigned above or below the cutoff, which varies the cost of getting certification, (2) having the option to retake below the cutoff, and (3) being induced by the encouragement effect to be more likely to retake the closer one scores to the left of the cutoff. Evaluating Assumption 4 therefore requires evaluating whether any of these steps can plausibly affect earnings through a channel other than affecting the likelihood of becoming a teacher.

By definition, the quasi-random nature of assignment around the cutoff means that whether one passes or fails near the cutoff is excluded from other possible factors that may influence earnings. The less obvious question is whether the process of *retaking* affects earnings through channels other than through becoming a teacher. Given that retaking mechanically increases the likelihood of qualifying for teacher certification, it is reasonable to assume that the main channel through which retaking affects earnings is through affecting the likelihood of becoming a teacher. That being said, there is one compelling story which, if true, would violate the exclusion restriction: if preparing to retake the exam led test-takers to gain additional motivation or human capital, retaking could increase their future earnings potential through channels unrelated to becoming a teacher. However, this story seems unlikely to be true given the short wait period for retaking and the content of the Praxis. The minimum, and modal, wait time is 28 days, and the entry Praxis exam tests middle-school math and reading skills. In light of this, it is reasonable to assume that the IV satisfies the exclusion restriction.

Estimation. Leveraging the lack of discontinuity and the kink in the likelihood of getting certified at the cutoff, I use a fuzzy RK design to estimate the effect of certification on earnings

via 2SLS:

$$\begin{aligned} \text{Structural eqn.} \quad Y_{i,t+4} &= \beta_0 + \beta_1 C_{i,t+4} + \beta_2 S_{it} + \epsilon_{it} \\ \text{First stage.} \quad C_{i,t+4} &= \alpha_0 + \alpha_1 (D_{it} \times S_{1i}) + \alpha_2 S_{it} + \eta_{it}, \end{aligned} \tag{1}$$

where $C_{i,t+4}$ is an indicator for whether one is certified within 4 years of their first attempt at the Praxis CASE, S_{it} is their first attempt score on the math portion of the Praxis CASE, and $D_{it} = \mathbb{I}(S_{1i} \geq 0)$. The main variable of interest is β_1/α_1 , a weighted average treatment effect of becoming certified on the outcome $Y_{i,t+4}$. The main outcomes of interest, $Y_{i,t+4}$, are earnings four years after the first test attempt, and whether one is employed four years after the first test attempt, particularly as a teacher.

Estimand interpretation. Like the RD estimand, the RK estimand can be interpreted as a weighted average treatment effect across the distribution of individual types (DiNardo and Lee, 2011). More specifically, I show that the RK estimator is a weighted average treatment effect for what I call “re-taking compliers”: individuals whose choice to retake is responsive to their first-attempt score. In other words, the weighted average does not include individuals who would never retake the test or who would always retake the test.

To provide the intuition behind this interpretation, I compare and contrast the local average treatment effect (LATE) interpretations of the RD and RK estimands. First, consider the imaginary world in which the Praxis is a one-shot exam. As shown in Figure II, a clear discontinuity arises in the first-stage relationship between first-attempt scores and whether one becomes a teacher, meaning whether one passes or fails near the cutoff can be used to instrument for whether one becomes a teacher in an RD design. In the LATE framework, the RD estimand gives a weighted average treatment effect for “compliers,” i.e., those who let the instrument dictate their receipt of the treatment. Compliers, in this world, are individuals whose first-attempt scores perfectly predict whether they become teachers. If they pass math on their first attempt, they go on to become teachers, but if they fail math on their first attempt, they never become teachers.

Now consider the real-world context in this paper, in which retaking is allowed and the first-stage relationship between first-attempt scores and the likelihood of becoming a teacher has a strong kink but no discontinuity, justifying the use of an RK design. Similarly to the RD, the RK estimand gives a weighted average treatment effect for a group akin to “compliers.” However, the traditional definition of a complier does not carry over to this setting because individuals can take multiple attempts at treatment. What does it mean for one to “comply”

with their original passing or failing status if retaking is an option? To characterize the sample included in the RK estimand, in Supplementary Materials I.C, I take the classical concepts of compliers/always-takers/never-takers/defiers and re-cast the definitions to account for retaking behavior. Combined with the identification proofs in Supplementary Materials I.A, I show that the RK estimand is a weighted average treatment effect for individuals whose choice to retake is responsive to their first-attempt score, whom I call “re-taking compliers.” This finding is intuitive: because the kink and lack of discontinuity in the first-stage is caused by the retaking process, it is sensible that the RK estimand is identified off of individuals whose decision to retake responds to their quasi-random assignment near the cutoff.

II.E. Regression Kink Estimates of the Pay Gap

First-stage and reduced form plots. Figure I shows the first-stage relationships between the first-attempt test scores and the likelihood of reaching any point of the teacher pipeline: ever passing the Praxis CASE, enrolling in an EPP, getting certified, and becoming a teacher. The main treatment of interest is getting certified, but for completeness, I show the progression of first-stage relationships for various outcomes along the pipeline.

At the earliest stage of the pipeline, there is a strong discontinuity at the cutoff in the likelihood that one ever passes math. However, the discontinuity shrinks to virtually zero as we progress along the pipeline. In its place, there emerges a kink at the cutoff in the likelihood one reaches any point of the pipeline.²⁶

I also show graphical evidence of the “reduced-form” relationship between the first-attempt test scores and the outcomes of interest—earnings and attachment to the labor force four years after attempting the Praxis for the first time—in Figure V. There appears to be no discontinuity or kink at the test score cutoff in the likelihood that individuals are employed, suggesting no effect of engaging in the teacher pipeline on being in the workforce. However, there is a weak kink at the cutoff in short-run earnings.

Treatment effect estimates. I now present my main estimates. I maintain a hands-above-the-table approach when it comes to bandwidth selection: I estimate the weighted average treatment effects for a wide range of bandwidths, ranging from an under-powered bandwidth of two up to a large bandwidth of thirty. I show the estimates at the optimal bandwidth

²⁶Due to our small sample sizes, the corresponding F-stats are small (Figure A.1).

recommended by [Calonico, Cattaneo, and Farrell \(2020\)](#) in Table II and for the full range of bandwidths in Figure VI.

I begin with the effect of passing the Praxis CASE and getting certified on the likelihood of employment. Leveraging the discontinuity in the pass rate at the cutoff, Table II, Panel A and Supplementary Figure A.2 show that passing the entire Praxis has no effect on the likelihood of short-run employment. Similarly, leveraging the kink in the likelihood of certification at the cutoff, Figure A.4 shows that getting certified has no effect on the likelihood of short-run employment. Given these null effects, in the following analysis of treatment effects on earnings, I restrict the sample to those who are employed (i.e., those with non-zero earnings).

Figure VI shows the treatment effect estimates of becoming certified and becoming a teacher on earnings across a range of bandwidths. Panels (a) and (b) show that the first stages begin to gain statistical power around a bandwidth of 12 to 14, coinciding with the MSE-optimal bandwidth recommended by [Calonico, Cattaneo, and Farrell \(2020\)](#), indicated by a vertical dotted line at 14.²⁷ The main RK estimates in Panels (c) and (d) suggest there is a large, positive effect of getting certified and becoming a teacher on earnings of around \$18,000–\$21,000. The estimates gain precision and are relatively stable across all bandwidths in Panel (d), suggesting that becoming a teacher is the main mechanism driving the overall effect on earnings.

Interpretation. The differences between the RD and RK estimates are intuitive given the context. The skills developed along the pipeline should have the most significant pay-off for those who complete their EPP training. Indeed, given that 87% of those who become certified also become teachers, finding employment as a teacher appears to be a key driver of the treatment effect of getting certified.

The RK results indicate that for “retaking compliers,” who let their first-attempt score dictate whether they retake the exam, there is a large positive premium of getting certified: around \$18,000–\$21,000 per year. Implicit in this interpretation is an assumption about monotonicity in the retaking behavior: I assume “retaking never-takers” do not exist—that is, people who would only retake the exam if they passed on their first attempt and would otherwise not retake the exam. The RK estimate also does not apply for “retaking always-takers”—that is, people who would retake the exam no matter what, even if they had passed on their first attempt.

²⁷Consequently, the estimates shown in Table II, Panels B and C are for a fixed bandwidth of fourteen, at which the estimates begin to stabilize.

This is reassuring, as “retaking always-takers” may be individuals with the largest expected return to becoming a teacher; including always-takers would thus inflate our estimates.

Robustness checks. Due to the size of KY and the proximity of many of its counties to borders of neighboring states, the KY state education department has an agreement with some neighboring states allowing teachers to carry their certification across state borders. To check whether these agreements may affect the results, I omit one college institution from my sample—Northern KY University—which is particularly known to send its graduates across KY’s border to work in Cincinnati, Ohio. The results are not sensitive to this omission.

In addition, because of small sample sizes and the resulting weak first-stages in the RK, I present the classic robust 2SLS standard errors as well as Anderson-Rubin confidence intervals (CIs), tF CIs resulting from implementing the [Lee et al. \(2022\)](#) tF procedure, and the VtF CIs from implementing the [Lee et al. \(2024\)](#) VtF procedure. The tF confidence intervals are by far the widest, encompassing zero for bandwidths less than fourteen. However, my preferred estimates using the VtF intervals closely correspond with the regular CI estimates and are narrower than the AR CIs.

The point estimates also do not change dramatically when I investigate the effect on longer-run earnings five or six years after taking the test, although the sample size becomes considerably smaller.

II.F. Empirical Strategy II: Event Study

The second strategy I use to examine the pay gap between teaching and teachers’ next-best options is an event study design. I examine how pay changes for teachers who exit teaching and change jobs using two data sources: the KY administrative data, and the Current Population Survey (CPS) matched sample. Relative to the RK, the event study permits me to “get away from the (test score) cutoff,” in that one would not expect that exiting teachers are necessarily those who would have scored near the passing cutoff on the Praxis. One caveat to this design however is that the resulting estimated pay changes may not be causal, since moves may be endogenous to pay. To address this concern, I apply the event study design to teacher exits that were likely driven by factors unrelated to individual teachers’ preferences, including “mass layoff” (school closure) events and principal turnover. These results provide complementary evidence to the RK estimates of the pay gap, using a different source of variation.

Event study specification. Using the administrative data, I identify the sample of teachers who move out of teaching at some point between 2009 and 2018, whose move cannot be attributed to retirement, and who are employed as a teacher for at least three years prior to the move. Aligning their dates of exit at $t = 0$, I estimate the following specification via OLS:

$$Y_{it} = \beta_0 + \sum_{\tau \in \{-2, 0, 1, 2\}} \alpha_{\tau} \cdot \mathbb{I}(t = \tau) + X_{i,t=-1} \gamma + \varepsilon_{it}, \quad (2)$$

where the coefficients α_{τ} capture the adjusted earnings difference between year $t = -1$ and year τ , and $X_{i,t=-1}$ are controls for the teachers' characteristics one year prior to exit.

I also estimate a two-period version of the estimating equation on the sample of individuals who are observed leaving teaching in the matched Current Population Survey (CPS). Studying wage changes in the CPS has two main advantages. First, it allows us to check whether the pay gap we estimate in the administrative data is due to features data-specific features, such as the coverage of only within-state employment. Second, the CPS contains data on occupations and hours across industries, allowing me to examine the types of occupations and part-time/full-time work that teachers leave teaching for.

II.G. Event Study Estimates of the Pay Gap

Figure VIII illustrates the raw data underlying the event study design: specifically, the earnings trajectories for the same sample of individuals who leave teaching in 2010, as well as the earnings trajectories for individuals who stay in teaching for the entire period of 2009 till 2019. The only group that continually earns more than they would have as teachers are those who change to other public school employment; these individuals mostly move to administrative positions, which offer higher pay scales than teaching. In contrast, those who move to employment outside of teaching earn on average \$21,000 less in their first year away from teaching. Their earnings recover slightly once they transition back into teaching. Those who leave to non-employment mostly do not return to teaching or other employment and thus earn the least.²⁸

The raw means, while illustrative, may reflect pay differences biased by different initial levels of pay across districts or different ages among leaving teachers. I therefore turn to Figure VII, which shows the estimated coefficients from the event study specification. I control for pre-determined observables at the time of exit including gender, race, age and years of experience

²⁸I code non-employment as zero earnings to maintain the same sample throughout.

teaching at the time of move, and district fixed effects.

The regression-adjusted estimates suggest that leaving teachers experience a pay decrease of between \$18,000 and \$25,000 a year upon leaving teaching. I also find suggestive evidence that the earnings drop is not purely driven by selection in worker transitions, in that the gap persists in cases where the separations are plausibly exogenous to individual-level unobservables. Although the data do not include reasons for exit, the data are rich enough such that I can (1) distinguish voluntary exits from involuntary ones, and (2) identify when exits coincide with a more systematic turnover event at a school. First, to investigate whether voluntary separations might be driving the gap, I examine the earnings trajectories separately for teachers who were tenured versus not tenured at exit.²⁹ Voluntary moves made by tenured teachers coincide with larger earnings drops than plausibly involuntary moves made by untenured teachers. However, even for untenured teachers, the earnings drop is still on the order of \$15,000 to \$25,000. Secondly, inspired by the mass layoffs literature, I study what happens to earnings for teachers whose move coincides with a “mass turnover” event at their school or a change in the school principal, which are less likely to be driven by individual-level factors.³⁰ I find similarly large earnings drops for teachers who leave from said schools.³¹

I draw similar conclusions using the matched CPS. From the matched CPS, I find that teachers leave to jobs that pay a median of \$13,000 less (or an average of \$5,000, due to some extreme positive outliers) per year than teaching. The most popular sectors (occupations) one year after exit include education (e.g., teacher assistants, administrators, childcare), healthcare (e.g., nurses, social workers, counselors), administrative (e.g., secretaries, administrative assistants, office clerks), and retail (e.g., sales worker supervisors, cashiers, waiters). Nearly half of leaving teachers exit to full-time jobs while one-fifth exits to part-time work.

Overall, the event study findings corroborate the evidence from the RK of a large, positive pay gap. The event study estimates are surprisingly similar to the RK estimates, and are also plausible in light of the occupations analysis from the CPS, given that part-time or even full-time work as a cashier or secretary may pay much less than teaching per year. I also find

²⁹Teachers enjoy a degree of job security after earning “tenure” (working for four years consecutively in the same district), which effectively protects them from being fired, making their moves largely voluntary.

³⁰I define a “mass turnover” as a school turnover rate that falls above the 75th percentile in a given year.

³¹These estimates are also similar in magnitude to the earning effects of mass layoffs and worker displacement found in the established labor literature. For instance, [Jacobson, LaLonde, and Sullivan \(1993\)](#), [Couch and Placzek \(2010\)](#), [Davis and Von Wachter \(2011\)](#), and [Lachowska, Mas, and Woodbury \(2020\)](#) find first-year earnings losses after worker displacement of between 33-66%, while I estimate first-year earnings losses of 33-40% among leaving teachers whose exit coincides with a school-level shock.

suggestive evidence that gap persists even ten years after the initial exit from teaching.

II.H. Contextualizing the Pay Gap Estimates

Using two empirical strategies that leverage different sources of variation in who is a teacher in KY, I find consistent evidence that the pay gap between teaching and teachers' next-best jobs ranges between \$18,000 to \$21,000 a year (in 2018 USD). Is this really the pay gap for the average teacher? And how does the pay gap compare to what teachers actually earn in KY?

I interpret this range of estimates as providing an *upper bound* on the pay gap for the average teacher. The RK treatment effects are weighted towards marginal test-passers, who include nearly 40% of the sample of test-takers in the optimal bandwidth around the cutoff. My empirical strategy thus gives no weight to individuals who would have passed the test regardless, which includes a sizable share of individuals who obtain the highest possible score of 200. Taking performance on a standardized test as a proxy for human capital, marginal and lower-scoring students may also be expected to have lower earnings potential; indeed, as shown in the reduced form analysis, short-term future income is weakly increasing in initial test scores. The test-takers driving the results are therefore those with relatively low earnings potential compared to the average teacher in the sample. Similarly, while I do not observe reasons for teacher exit, one possible reason especially pre-tenure is poor performance. While it remains an open question whether teacher value-added or other measures of teacher effectiveness correlate with general ability (Chingos and West, 2012), it is possible that the event study estimates are being driven by lower general ability individuals. As such, both designs deliver the pay gap between teaching and the next-best jobs of individuals with potentially low earnings potential, leading us to conclude that the estimated pay gaps serve as an upper bound. This is advantageous in our context: given that the primary goal of this paper is to assess the extent to which teachers earn rents, an upper bound estimate is more conservative and therefore preferred to a lower bound estimate of the pay gap.³²

Finally, I use a back-of-the-envelope calculation to put my estimates in context. Table III shows the results of dividing the estimates of the pay gap by benchmark values of KY teachers' annual earnings. For the RK estimation, where the main outcome is earnings two

³²It is also worth mentioning that although these estimates provide insight into the pay gap at different points in the teaching pipeline, one may be interested instead in the gap in lifetime earnings between teaching and teachers' next-best jobs. Because earnings growth is lower in teaching than in other jobs and industries, it is plausible that even with a large early-career pay gap, there may be a small or even negative lifetime earnings gap between teaching and other occupations.

years after college, I use the average teaching salary across districts for those with two years of experience and the most common set of credentials as a benchmark: approximately \$45,000 in 2018 USD.³³,³⁴ For the event study analysis, where the main outcome is earnings one year after leaving teaching, I use the average teaching salary one year prior to exit as a benchmark: around \$55,000 in 2018 USD.³⁵ Column (2) shows my preferred calculation, which suggests that an upper bound estimate of the pay gap is equivalent to between 33–40% of a KY teacher’s annual earnings, depending on the benchmarks used.³⁶

III. ESTIMATING THE WORKING CONDITIONS GAP

III.A. Identification Challenge

In the second half of this paper, the goal is to estimate how much more or less teachers value the non-wage amenities in teaching relative to the amenities in their next-best job options. Doing this requires not only estimating teachers’ willingness to pay for various amenities, but also knowing what teachers’ next-best jobs are and how the amenities differ between teaching and teachers’ next-best jobs. Both are challenging to pin down using observational data on workers’ job transitions, and data on amenities are generally scarce.

To learn about the amenities in teachers’ next-best jobs, I leverage survey data from the American Working Conditions Survey (AWCS) (Maestas et al., 2023) and the results of the earlier quasi-experimental analysis. The RK design and event studies show that teachers’ next-best jobs are low-paying options in the education, healthcare, administration, and retail industries. Reasonable proxies for the amenities in teachers’ next-best job can therefore be obtained using the survey responses of workers in the AWCS in the education and healthcare sectors earning below-median wages.

To estimate teachers’ preferences for a wide range of job attributes, I administer a set of

³³These credentials can be thought of as courses that can be obtained within the first two years of teaching.

³⁴There is essentially no variation in these average salaries between 2018 and 2022 after adjusting for inflation. The \$45,000 salary serves as a lower bound on one’s actual annual earnings: taking up additional responsibilities around the school can generate additional pay based on another fixed salary scale, and it is common for teachers to work a second job.

³⁵Based on the salary schedules, this implies that the average leaving teacher has between 5–10 years of experience.

³⁶My preferred estimate is the effect of actually working as a teacher on earnings, which at the optimal bandwidth falls around \$18,000, and is virtually stable over the surrounding bandwidths as seen in Figure VI. For transparency, we also show our calculations our estimate of the effect of becoming certified to teach on earnings, which at the optimal bandwidth falls at around \$21,000. However, as Figure VI shows, this point estimate shrinks and appears to stabilize at lower values as the bandwidth and first-stage F-stat grows.

choice experiments to a representative sample of teachers in KY. Teachers are presented with a series of job pairs with randomly varied job attributes and asked which job they most prefer. The resulting choice data allows me to observe the trade-offs teachers face and the choices they make, and back out teachers’ willingness to pay for each attribute included in the experiment.

The following sections detail the survey design, estimation strategy, and results.

III.B. New Data on Working Conditions

Survey of KY teachers. To gain an understanding of how working conditions vary between teaching and other jobs, I field an online survey on a representative (once weighted) sample of teachers in the state of KY. The final sample include responses from over 1200 teachers, and a total of 3,700 choices made in the choice experiments.³⁷ The email response rate was 6%, exactly the average email survey response rate according to Qualtrics.

The two goals of the survey are to elicit information about teachers’ working conditions, and to run a choice experiment that allows me to estimate teachers’ willingness-to-pay for various working conditions. To achieve these goals, I randomize school districts into two groups and give school staff in each group a survey that closely replicates the American Working Conditions Survey (AWCS). The first group received a survey with the exact same questions and experimental design as the AWCS (Maestas et al., 2023).³⁸ The second group received a survey that asked about different, “teaching-specific” working conditions, but retained the same structure and experimental design. The teaching-specific working conditions included: job hours that follow the school calendar (including “summers off”), hours that allow one to leave the workplace mid-day, working with children, working with parents (“clients”), job security, support from management in situations of conflict with clients, facing hostility at work (e.g., verbal aggression), and the amount of respect the job receives from the public. I selected attributes that were either reported reasons for dissatisfaction in teaching (e.g., school safety, student discipline, administration) in the Teacher Follow-up Survey (NCES), or that were salient

³⁷Data come from two survey rounds run in May 2024 and another two rounds in May 2025. Each respondent was asked to respond to a minimum of three job choices, with the option to evaluate one or two additional choices. The majority chose to evaluate the maximum of five job choices.

³⁸The AWCS was the first survey designed to elicit “detailed information about a broad range of working conditions in the American workplace,” and was first fielded to a nationally representative sample of U.S. workers in 2015 by Maestas et al. (2023) through the RAND American Life Panel (ALP). The survey contained two parts: a questionnaire asking respondents about their current job attributes, and a stated-preference module that presents respondents with ten pairs of hypothetical jobs and asks which job they prefer in each case.

characteristics specific to teaching (e.g., school calendar, working with children).^{39,40}

There are distinct advantages to surveying KY teachers in this way. Targeting KY teachers ensures internal consistency, meaning the willingness-to-pay estimates and earlier pay gap estimates are derived from the same population. Asking respondents about the same core attributes studied in the AWCS ensures I am assessing job attributes that workers find salient when choosing between jobs. Finally, given my survey provides the first set of estimates of teachers’ willingness-to-pay for a wide range of general job attributes, it is useful that the AWCS data provides several natural benchmark estimates: namely, willingness-to-pay estimates for women, college-educated workers, and workers in the education and health sector.⁴¹

Table IV shows summary statistics on the demographic composition of the survey respondents, with comparisons to the public school teaching population in KY. The survey sample closely resembles the teaching population in terms of race (95% white), gender (78% women), and likelihood of working at a Title I school (67–71%). However, older teachers and teachers with more teaching experience are over-represented in the survey. Due to these differences, I construct survey weights for the survey respondents based on the administrative population data and reweight the respondents in all following analyses.

III.C. Incidence of Working Conditions in Teaching

I begin by examining the incidence of the nine core non-wage job attributes among KY teachers. Figure IX presents a summary of the incidence of working conditions for four groups of workers: KY teachers, all workers in the U.S., college-educated workers, and workers in the education

³⁹The inclusion of “public respect” as a job attribute was inspired by Dube, Naidu, and Reich (2025). The Teacher Follow-Up Survey is part of the Schools and Staffing Survey run by the National Center for Education Statistics (NCES). The teaching-specific questions were formulated by adapting existing questions on similar topics from the AWCS, the Centers for Disease Prevention (CDC) National Institute for Occupational Safety and Health (NIOSH) Quality of Worklife Module, and the Pew Research Center’s American Trends Panel.

⁴⁰Notably, this list excludes retirement plans. This is intentional. In KY, teachers are offered a defined benefit plan, provided by the Teachers’ Retirement System of Kentucky, but no social security benefits—and teachers’ WTP for retirement plans has been previously explored, (Johnston, 2025). Furthermore, with our data, it is difficult to tell whether teachers’ next-best jobs offer retirement benefits or not, and whether the provision of these benefits is a relevant job attribute to consider for most workers: young teachers, who have the highest turnover rates in the profession, likely have relatively low willingness to pay for retirement plans, as found by Johnston (2025).

⁴¹The AWCS data also gives an imprecise benchmark for teachers—“imprecise” because since the AWCS does not collect occupation for the majority of respondents, only a small sample of 46 (out of the total 1,748 respondents) report a string similar to “teacher” as their occupation (e.g., “teach,” “teac”), and these individuals reside in various places across the country.

and health industries with below median pay.⁴² Means are weighted using survey weights.

Compared to the job of the average college-educated worker, teaching is significantly less likely to offer the valued working conditions included in the AWCS, like setting one’s own schedule, but is much more likely to offer “teaching-specific” attributes, like working with children. Most college-educated workers report being able to set their own schedule (67%), being able to telecommute (54%), and having more than 15 days of paid time off (68%). In contrast, among teachers, only a small share report having some scheduling flexibility (12%), being able to telecommute (4%), or having more than 15 days of paid time off (5%). Teachers are also more likely to report more physically demanding work (2% report mostly sitting, compared to 58% of college-educated workers). Predictably however, teaching is highly likely to offer “teaching-specific” job attributes. All teachers report having hours that follow the school calendar and working with children, and the majority report having high job security.

These differences may not be surprising given the traditional model of classroom teaching and the structure of the school year: classroom teachers are required to be physically present at school five days a week, and “summers off” do not technically count as paid time off. However, teaching also differs from other jobs along other less obvious dimensions. Perhaps most starkly, teachers are significantly more likely than all of the other groups of workers to experience a stressful “fast-pace” at work (87% vs. 64% for college-educated, 67% for women, and 70% for all workers). Relatedly, teaching is significantly more likely to involve dealing with hostility in the workplace compared to other jobs. The teaching job is also less likely to offer training opportunities to develop skills that will transfer to other jobs (53% in teaching vs. 74% for college-educated workers), less likely to give workers autonomy at work (81% vs. 91%), and more likely to require working alone (41% vs. 33%). In fact, the only positively-valued amenity from the AWCS that teaching is significantly more likely to offer is the opportunity to make a positive impact on one’s community or society (64% in teaching vs. 39% for jobs of college-educated workers).

The data on the small sample of teachers in the AWCS corroborate my survey findings. Using the AWCS data, I construct similar comparisons between other workers and teachers, where I identify teacher respondents by merging in occupation information from the RAND American Life Panel.⁴³ Reassuringly, I reach the same qualitative conclusions using the AWCS

⁴²All statistics in the figure, along with additional summary statistics on the job attributes in teaching compared to the jobs held by other workers in the AWCS, are provided in Supplementary Tables A.1, A.2, and A.3.

⁴³Supplementary Table A.4 shows summary statistics on the differences between the AWCS teachers and

data: that is, that teaching is less likely to offer valued working conditions compared to the jobs of observationally similar workers. Both surveys also show that teaching pays less than the jobs of demographically similar workers, in line with the findings of previous studies ([Allegretto and Mishel, 2018](#)).⁴⁴

Finally, I focus on the comparison most relevant to the working conditions gap: how the working conditions in teaching compare to those of teachers’ “next-best” jobs. The RK and event studies suggest that the average teachers’ next-best job is a low-paying option in the education or healthcare sector.⁴⁵ I therefore compare the attributes in teaching to jobs in the AWCS in the education and healthcare sectors with below-median pay, represented by the triangles in Figure IX.⁴⁶ The differences in attributes all go in the same direction as the differences between teaching and college-educated workers’ jobs: even when compared to their lower-paying next-best options, teaching is significantly less likely to offer most desirable attributes, aside from the “teaching-specific” conditions.

III.D. Estimating Teachers’ Willingness to Pay for Working Conditions

To estimate teachers’ willingness to pay for each job attribute, I conducted a series of stated-preference experiments with each survey respondent. I detail my experimental design and estimation strategy below, with additional details in Supplementary Materials I.D. I include “trick” questions to address the concern with that survey respondents may only skim the questions and thus provide inaccurate responses. My results are robust to excluding inattentive respondents.

other workers. Statistical tests in Supplementary Tables A.5 and A.6.

⁴⁴Supplementary Table A.1 shows that KY teachers are contractually obligated to work a similar number of hours per week as the average U.S. worker, women worker, or college educated worker in the AWCS. However, the hourly wage in teaching is closer to the wage of the average U.S. worker or woman worker than it is to the average college educated worker.

⁴⁵The next-best jobs identified using the quasi-experimental designs align with teachers’ beliefs, as revealed in the survey. Figure A.6 shows the the distribution of responses to the question: what industry they would work in if not teaching? Over a third of teachers indicated that their next-best job would be in either the education or healthcare sector. The next most popular industries included professional and business services (12.5%), leisure and hospitality (11.5%), government (6.8%), and finance, trade, and information.

⁴⁶Supplementary Table A.2 summarizes the job attributes in teaching versus in the average job in the next-best industries, while Supplementary Table A.3 summarizes the attributes of the jobs that pay below the industry median wage. The distribution of attributes is similar between the average paying and the below-median paying jobs within industry. In terms of pay, the average job in all industries, aside from leisure and hospitality, pays more on average than teaching does. In terms of working conditions, the average job in all other industries is substantially more likely to offer many of the desirable job attributes that teaching does not have, including scheduling flexibility, telecommuting options, relaxed pace of work, and more than 15 days paid time off.

Experimental design. In the first part of the survey, respondents are asked about the pay, hours, and working conditions in their current/former teaching job. Respondents are randomly assigned one of two sets of questions. Half the sample is asked about the nine core job attributes included in the AWCS: schedule flexibility, telecommuting opportunities, physical demands, pace of work, autonomy, paid time off, working with others, job training opportunities, and impact on society.⁴⁷ The other half is asked about nine “teaching-specific” job attributes that were not included in the AWCS, such as having “summers off,” working with children, and dealing with parents. To obtain each respondent’s hourly wage, I ask respondents to report their teaching earnings (at the hourly, weekly, biweekly, twice-monthly, monthly, or annual level), their hours worked per week, and their weeks worked per year.

In the second part of the survey, respondents are presented with three choice experiments.⁴⁸ In each experiment, respondents are asked to select between two hypothetical job profiles that partially vary in terms of job attributes, hours, and wages, exactly following the technical and visual design described in [Maestas et al. \(2023\)](#).

I construct the hypothetical job profiles in two stages. First, I create the respondent’s “baseline job,” which functions as an initial set of values that can be varied to create hypothetical job profiles. To keep the hypothetical job choices familiar to the respondent, the baseline job is defined as the pay, hours, and job attributes of the respondent’s current/former teaching job that they reported on in the first part of the survey.⁴⁹

Second, I create hypothetical job profiles Job A and Job B by taking the baseline job and randomly selecting two of the nine non-wage job attributes (without replacement) to vary. I also vary the offered wage and hours per week between the two jobs, such that in total, each job is characterized by eleven total characteristics. Hours are randomly varied to be one of 5-hour intervals between 15 and 60 hours per week. Wages are generated to be similar on average to the respondent’s actual hourly wage. Given the respondent’s actual hourly wage w_i , the hypothetical wages for Job A and Job B are given by $\theta_A w_i$, where $\theta_A \sim N(1, 0.1^2)$ and

⁴⁷See Supplementary Table 2 of [Maestas et al. \(2023\)](#) for a list of the job attributes and their potential values.

⁴⁸To maximize the data collected without inducing survey fatigue, I also gave respondents the option to respond to at most two “bonus” experiments, so that respondents could complete up to five experiments. No additional incentives were provided; survey respondents were simply told that completing the additional experiments would “be a huge help to the study.” While the majority actually did respond to at least one bonus experiment, my main results use the first three experiments to avoid selection bias from additional responses.

⁴⁹However, to address concerns that respondents may prefer job profiles that they are familiar with, which could bias the willingness-to-pay estimates, I also define a “common” baseline job—essentially the average teaching job based on the survey pilot—and use it in one of the three experiments. The values of the common baseline job are shown in Supplementary Table 2 of [Maestas et al. \(2023\)](#). I find that the inclusion of the common baseline jobs does not affect our willingness-to-pay estimates.

$\theta_B w_i$, where $\theta_B \sim N(1, 0.1^2)$ respectively. To ensure that the wage values do not stray too far from the respondent’s reported wage, I truncate θ_A and θ_B at 0.75 and 1.25. To make the hypothetical wages interpretable for the respondent, I display the earnings for each hypothetical job in terms of the units the respondent used to report their earnings. Finally, to maximize estimate precision, I redraw wages and/or attribute values in cases where one job dominates the other on all dimensions.⁵⁰

Following the above steps produces a Job A and a Job B that are identical in every way except for two non-wage attributes, the hourly wage, and the hours worked per week. Respondents are presented with a table showing a side-by-side comparison of the two jobs’ attributes, and are asked whether they “Prefer Job A,” “Prefer Job B,” “Strongly Prefer Job A,” or “Strongly Prefer Job B.” My presentation of the job pairs, shown in Supplementary Figure A.5, is virtually identical to the presentation in Supplementary Figure 3 of [Maestas et al. \(2023\)](#).

Estimation. Using the choice data from the choice experiments, I estimate KY teachers’ willingness-to-pay for each job attribute. Assume that the respondents have the following indirect utility function:

$$V_{ij} = \alpha + A'_{ij}\beta_i + \delta_i \ln w_{ij} + \varepsilon_{ij}, \quad (3)$$

where A_{ij} is the vector of job attributes offered by job j and w_{ij} is the corresponding wage. Under the assumption that ε_{ij} is i.i.d. EVI, the likelihood that worker i chooses teaching over job k is given by the expression

$$\Pr(V_{it} > V_{ik}) = \frac{\exp[(A'_{it} - A'_{ik})\beta_i + \delta_i(\ln w_{it} - \ln w_{ik})]}{1 + \exp[(A'_{it} - A'_{ik})\beta_i + \delta_i(\ln w_{it} - \ln w_{ik})]}. \quad (4)$$

In theory, the β_i and δ_i subscripts allow for heterogeneity in preferences over amenities and pay. However, because I focus on workers in a specific occupation, and thus a specific and less variable group of individuals, I assume that $\beta_i = \beta$ and $\delta_i = \delta$ and use a standard logit model to estimate β and δ . I aggregate the four possible responses into a single dummy variable that indicates whether the workers prefer Job A or not, and use survey weights in estimation. I also conduct a battery of robustness checks, including: (i) excluding “inattentive” respondents, i.e. those who incorrectly answered both trick questions, (ii) estimating a probit model instead of

⁵⁰See Supplementary Materials [I.D](#) for details on randomization.

a logit model to evaluate sensitivity in the error term distributional assumption, (iii) excluding experiments that used the “common” baseline job instead of the respondent’s own baseline job, (iv) estimating the model without survey weights. Despite my relatively small sample compared to the AWCS, I find that my willingness-to-pay estimates are robust to these variations.

I transform the estimated parameters from the discrete choice model into measures of the teacher’s willingness-to-pay for each desirable job attribute r . If teacher i is willing to pay WTP_i^r for an attribute r , then they should be indifferent between a job that offers w_i without r and a job that offers r but a lower wage, $w_i - \text{WTP}_i^r$:

$$\delta \ln w_i = \beta^r + \delta \ln[w_i - \text{WTP}_i^r], \quad (5)$$

where β^r denotes the average teacher’s marginal utility from attribute r and δ is the average teacher’s marginal utility of the log wage. Rearranging gives the object of interest:

$$\text{WTP}_i^r = w_i \left[1 - \exp \left(\frac{-\beta^r}{\delta} \right) \right]. \quad (6)$$

For ease of interpretation, I report the estimates in terms of $1 - \exp \left(\frac{-\beta^r}{\delta} \right)$, so that gaining attribute r in one’s job is equivalent to a $100 \left[1 - \exp \left(\frac{-\beta^r}{\delta} \right) \right] \%$ wage increase.

Comparison between jobs. Finally, I use the preference estimates to estimate how much teachers are willing to pay for one job over another. I study three such comparisons: (1) the average teacher’s total valuation for the “best” job over the “worst” job (i.e., the set of all desirable attributes over the set of none of them), (2) the average teacher’s total valuation of all the desirable amenities teaching does not offer, and (3) the total amenity value the average teacher would get from switching from teaching to the their next-best job.⁵¹

I introduce some notation that will be used throughout the definitions for each of the three measures below. Let S denote the set of all attributes. Let $\beta^{s,1}$ denote the preference parameter for the highest attribute value of s . In cases where there are more than two values for attribute s , let $\beta^{s,2}$ denote the preference parameter for the second highest attribute value of s .^{52,53}

The first measure is teacher i ’s willingness to pay for the “best” job over the “worst” job,

⁵¹I continue using low-paying jobs in the education and health sectors in the AWCS as a proxy for teachers’ next-best jobs.

⁵²If the attribute has only two values, $\beta^{s,2}$ is 0.

⁵³For example, if s is the physical demands attribute, $\beta^{s,1}$ is the preference parameter of moderate physical demands (the most preferred value) and $\beta^{s,2}$ is the preference parameter for mostly sitting.

defined as:

$$\text{WTP}_i^{\text{FULL}} = w_i \left[1 - \exp \left(\frac{-\sum_{s=1}^S \beta^{s,1}}{\delta} \right) \right]. \quad (7)$$

The sum in this expression only puts positive weight on the estimated preferences for the most preferred value of each attribute.⁵⁴ This measure is equivalent to the best-to-worst measure used in [Maestas et al. \(2023\)](#).

The second measure is teacher i 's willingness to pay for all of the desirable amenities that teaching does not offer:

$$\text{WTP}_i^{\text{BEST}} = w_i \left[1 - \exp \left(\frac{-\sum_{s=1}^S (\beta^{s,1} - \beta^{s,2} \cdot \mathbb{I}_{\text{teach}}^{s,2})(1 - \mathbb{I}_{\text{teach}}^{s,1})}{\delta} \right) \right], \quad (8)$$

where $\mathbb{I}_{\text{teach}}^{s,j}$ is an indicator variable for whether the average KY teaching job offers the j th highest attribute value of s or not. The summation only gives weight to the attributes that teaching does *not* offer. Thus, unlike Equation (7), this measure accounts for differences between the “best” job and a job that offers *some* desirable amenities, rather than none at all.⁵⁵

Finally, the third measure captures the compensating differential between teaching and teachers' next-best jobs. I define the measure as the respondents' willingness to pay for their next-best job—that is, a below-median-pay job in the education or healthcare sector—over teaching:

$$\text{WTP}_i^{\text{NXBEST}} = w_i \left[1 - \exp \left(\frac{-\sum_{s=1}^S (\beta^{s,1} - \beta^{s,2} \cdot \mathbb{I}_{\text{teach}}^{s,2})(\mathbb{I}_{\text{nxbest}}^{s,1} - \mathbb{I}_{\text{teach}}^{s,1})}{\delta} \right) \right], \quad (9)$$

where $\mathbb{I}_{\text{nxbest}}^{s,1}$ is an indicator variable for whether the average next-best job has the highest value of attribute s or not. The sum in this expression puts positive weight on β^r if r is offered by the next-best job but not by teaching, no weight on β^r if r is offered in both or neither of the jobs, and negative weight on β^r if r is offered by teaching but not by the next-best job.

For all three summary measures, I compute standard errors using the delta method and

⁵⁴Note that this baseline measure does not “double count” attribute values, which would inflate the measure. If an attribute has more than two values, I only use the coefficient associated with the attribute value with the highest willingness-to-pay estimate.

⁵⁵The expression also makes clear why it is necessary to include multiple attribute values, not only the highest valued ones, when valuing a job relative to teaching. Take for example the paid time off attribute, which has three values—no time, 10 days, and 20 days—that workers value in increasing order. If teaching offers 10 days paid time off (meaning $\mathbb{I}_{\text{teach}}^{s,2} = 1$), then when evaluating how much teachers would value having 10 additional days paid time off, we would want to include teachers' valuations of 20 days paid time off *netting out* their valuation of the 10 days paid time off that they already get as teachers, which is given by $(\beta^{s,1} - \beta^{s,2} \cdot \mathbb{I}_{\text{teach}}^{s,2})$.

cluster by respondent.

The willingness-to-pay parameters are identified so long as the respondents believed that the jobs in each pair, randomly varied characteristics aside, were otherwise identical. As such, I assume that respondents followed the explicit instruction in the survey to assume that the pair of jobs shown are identical in every way, including in ways that are not explicitly specified in the table. I also assume that there were no systematic differences in how respondents rated the subjective amenities like heavy versus moderate physical activity, which is reasonable given that teachers are somewhat homogeneous (e.g., mostly women) and engage in similar work.

III.E. Main Estimates of Teachers' Willingness to Pay for Working Conditions

I present my main estimates of KY teachers' willingness-to-pay for each job attribute in Figure X, alongside willingness-to-pay estimates for other workers from Maestas et al. (2023) and my own survey where available.⁵⁶ Each point can be interpreted as the percentage wage increase (or decrease) that a teacher would need to compensate for removing (or adding) a given job attribute, relative to some baseline attribute. For example, the estimate for teachers in the first row indicates that the average KY teacher is willing to pay 11.9 percent of their salary to be able to mostly sit at a job rather than engage in heavy physical activity.

I highlight four key findings. First, like other workers, KY teachers are willing to pay large amounts for certain working conditions. The most valued job attribute is paid time off: switching from a job with no paid time off to one with 20 days of paid time off is equivalent to a 26.8 percent wage increase. Teachers also highly value switching to a job with more boss/management support (equivalent to a 21 percent wage increase), with high job security (equivalent to a 24.5 percent wage increase), and with a schedule that follows the school calendar (equivalent to a 22.4 percent wage increase).

Second, I find that the average teacher and the average worker share similar preferences for most of the core AWCS attributes. The only exceptional difference is having a relaxed pace at work: teachers value switching from a fast-paced job to a relaxed pace job at a 10 percent wage increase, nearly twice as much as the average worker does.⁵⁷ One explanation for this difference

⁵⁶Estimates are also shown in Supplementary Table A.7.

⁵⁷Some of the insignificant differences in preferences are noteworthy as well. In particular, teachers and other workers share the same willingness to pay for having frequent opportunities to impact the community or society with one's work. This finding may be surprising given that, as shown in Figure IX, teaching is significantly more likely than other jobs to offer such opportunities. These estimates suggest that teachers are not strongly selected on community-impact-related altruism.

is the prevalence of teacher “burnout”: workers who experience fast-paced, stressful jobs at the time of the survey may place a higher value on a relaxed pace of work. Indeed, surveys show that stress and burnout are common among teachers and contribute to high turnover rates, especially among starting teachers (e.g., [Herman, Hickmon-Rosa, and Reinke \(2018\)](#)).

However, my third finding is that teachers’ preferences deviate from other workers’ preferences when it comes to teaching-specific attributes. Compared to other workers, teachers are more willing to pay for a job that follows the school calendar, involves working with children, and offers a high level of support from their boss/management (i.e., principal or school administrators) in conflicts with ‘clients’ (i.e., parents).⁵⁸ Furthermore, teachers are not willing to pay as much as other workers to work with ‘clients’ (i.e., parents) or job security.

Finally, I show that teachers are willing to pay a substantial amount of their salary to switch to typical their next-best job for its better amenities. Figure [XI](#) shows the estimates of how much teachers are willing to pay to switch jobs. Switching from the least desirable job to the most desirable job, in terms of AWCS amenities only, is equivalent to a 62.2 percent wage increase for KY teachers. Reassuringly, this estimate is very similar to the estimates in [Maestas et al. \(2023\)](#) of the value of switching from the “worst” job to the “best” job for the average American worker (a 55 percent wage increase) or for the average college-educated worker (a 60 percent wage increase). However, once teaching-specific attributes are included as well, switching from the “worst” job to the “best” job is equivalent to a much larger 93 percent wage increase. Switching from teaching to the “best” job, in terms of amenity value, is equivalent to a 43 percent wage increase. Finally, switching from teaching to the average teachers’ next-best job, in terms of amenities, is equivalent to a 17 percent wage increase.

Importantly, there is significant heterogeneity underlying the average willingness to pay estimates. The average total willingness-to-pay to switch from teaching to the next-best job of 17% is entirely driven by teachers with more than five years of experience. In contrast, less experienced teachers are willing to pay significantly more—as much as 51 percent of their teaching salary—to switch to their next-best job. This heterogeneity is mostly driven by differences in working conditions. Compared to experienced teachers, inexperienced teachers are significantly more likely to report facing hostility, less likely to receive support from management in situations of conflict with parents/children, and less likely to report high job security. The difference in job security in particular reflects the institutional setting: in KY, teachers are on

⁵⁸In fact, working with children is the only attribute included in the survey that other workers find to be a *disamenity*, while the average teacher has a positive willingness to pay to work with children.

one-year contracts until they obtain “tenure” (high job security), which can only be obtained after working in the same district for four consecutive years.

IV. DISCUSSION

Applying independent quasi-experimental designs to novel administrative data from KY, I find that teaching offers a large pay premium relative to teachers’ next-best options. I estimate a pay gap at between 33–40% of the average teachers’ annual earnings. I also find that KY teachers are willing to pay 17% of their salary to switch from the bundle of attributes in teaching to the overall more desirable bundle in their next-best jobs. The working conditions gap estimates vary by teaching experience, with inexperienced teachers willing to pay much more (51% of their salary) for the attributes in their next-best job than experienced teachers.

Combining my most conservative estimates to evaluate the teaching rent, given by

$$\text{Rent} \approx \underbrace{(\text{Pay}_T - \text{Pay}_{NB})}_{\text{Pay gap}} + \underbrace{(\text{WorkConditions}_T - \text{WorkConditions}_{NB})}_{\text{Working conditions gap}},$$

I find that experienced teachers in KY earn a rent of around 16% of their salary and a compensating differential of around 17% of their salary. In contrast, I find that inexperienced teachers in KY earn no rent, or even a “negative rent,” given their large working conditions gap.

My results suggest that, although teaching pays more than teachers’ next-best jobs, much of the pay premium is a compensating differential rather than a rent. In particular, for inexperienced teachers, the entire pay premium functions as a compensating differential due to the more challenging working conditions they tend to face. Through the lens of the compensating differentials framework in [Rosen \(1986\)](#), these estimates can be interpreted as showing that experienced teachers are inframarginal while inexperienced teachers are “marginal” to teaching. This is a sensible conclusion, as workers may derive rents from experience, and turnover rates are highest among less experienced teachers.

This conclusion is in line with increasing reports of teacher shortages and difficult working conditions in teaching, especially since COVID-19.⁵⁹, and has significant implications for education policy. Given that over \$500 billion are spent each year on teacher salaries, under the simple approximation that 70% of the wage bill is spent on experienced teachers’ salaries, my estimates imply that over \$120 billion are being spent each year to compensate teachers for

⁵⁹For example, see Sejla Rizvic, “Teachers, Facing Increased Level of Stress, Are Burned Out,” *The New York Times*, March 13, 2023.

challenging working conditions. The magnitude of this estimate raises the question of whether there may be more cost-effective solutions that could address the gap in working conditions between teaching and other jobs.

Assessing magnitudes. My estimates of the gap in the value of working conditions are large, especially for inexperienced teachers. However, a simple comparison of teacher pay in private versus public schools suggests that the magnitudes I estimate are plausible. Private school teaching jobs are the most similar jobs to public school teaching jobs in terms of the structure of the workday, the tasks involved, and the skills required. But private schools are also able to select students to enroll, leading to smaller class sizes and fewer behavioral issues.⁶⁰ Consistent with compensating differentials, in 2020–21, the average private school teacher in the US earned around \$19,000 *less* than the average public school teacher in 2018 USD ([National Center for Education Statistics, 2024](#))—around 42% of an early-career teaching salary in KY, a very similar number to my estimated working conditions gap of 51% for inexperienced teachers. While only suggestive, the private/public school pay gap is consistent with my finding that public school teachers may require a large compensating differential to make up for challenging working conditions driven by student behavioral issues and stress.

My estimates of the pay gap are also large. However, in light of the private/public school pay gap highlighted above, the magnitudes are plausible, especially when the point estimates are interpreted as an upper bound (as discussed in Section II.H). One may still be concerned that the estimates are biased due to the administrative earnings data coming from a state UI system that does not track moves out of state. For example, if the local opportunities for exiting teachers are lower-paying than teaching but the out-of-state opportunities for exiting teachers are higher-paying than teaching, then using UI data on wages may lead to overestimates of the pay gap. However, my estimates are in line with previous studies that have used event study designs to examine whether teachers leave teaching to higher or lower pay in multi-state data, including the NLSY and the LEHD (e.g., [Champion, Mastri, and Shaw \(2011\)](#); [Scafidi, Sjoquist, and Stinebrickner \(2006\)](#); [Goldhaber et al. \(2022\)](#)). These papers find similarly sized declines in pay following exits from teaching.

⁶⁰For example, see Ben Orlin, “Why Are Private-School Teachers Paid Less Than Public-School Teachers?”, *The Atlantic*, October 24, 2013 .

Qualitative evidence. The large pay gap estimates are internally consistent with KY teachers’ views expressed in the open-ended portion of the survey portrayed in Figure XII. When asked if there was anything more the teachers wanted to share about their experience choosing between teaching and other jobs, the most commonly raised issues were [student] behavior and parent involvement rather than pay.

Comparison to prior work. These conclusions may initially seem at odds with prior work that suggest teachers extract sizable rents. However, once differences in context and approach are taken into account, my findings can be interpreted as consistent with prior work. Many of the papers on rent extraction in the public sector focus on the role of unions in driving up rents. While teacher unions are known to be politically strong and active when it comes to collective bargaining, teacher unions are not as prevalent in the U.S. South: KY public school teachers had the 16th lowest unionization rate in the country in 2008 (58.6%).^{61,62} In 2011, less than 30% of KY teachers were covered by a union that actively participated in collective bargaining.⁶³ The low union coverage partly explains why I find evidence of moderate to low rents in an occupation that typically has strong union representation: indeed, the 16% rent I find for experienced teachers is very similar in magnitude to recent estimates of worker rents in the entire labor market—Lamadon, Mogstad, and Setzler (2022) estimate worker rents of 13-18%, while Jäger et al. (2024) similarly find worker rents of 10-14%—where most workers are not unionized. My findings are therefore most applicable to the many states with similar degrees of teacher union involvement, but not necessarily applicable to the average U.S. state.

This is also the first study to explicitly incorporate the value of non-wage amenities when evaluating whether workers extract rents. Most prior work uses total funding inputs for staff expenses to study rent extraction in teaching. However, most of the job attributes that I consider in our survey do not necessarily entail an explicit cost (e.g., working in a team versus working alone), and thus the presence or lack of such attributes would not be included in such metrics. My conclusions are consistent with a robust education literature that documents that poor working conditions are a key predictor of teacher turnover (e.g., Loeb, Darling-Hammond, and Luczak (2005); Borman and Dowling (2008); Ladd (2011); Boyd et al. (2011); Johnson,

⁶¹The fifteen states with unionization rates lower than KY, in decreasing order, were: Louisiana, Oklahoma, Florida, Georgia, Kansas, South Dakota, Wyoming, Virginia, North Carolina, Arizona, New Mexico, Mississippi, Arkansas, and South Carolina. The lowest rate of unionization was 26.9% in South Carolina.

⁶²Data from the School and Staffing Survey (National Center for Education Statistics, 2008).

⁶³See the report prepared by the KY Legislative Research Commission (Kentucky Legislative Research Commission, 2022).

Kraft, and Papay (2012)).

Finally, this paper provides new perspective on a few established findings about teachers. The first is that teachers have become increasingly negatively selected on academic performance (e.g., Bacolod (2007); Corcoran, Evans, and Schwab (2004); Podgursky, Monroe, and Watson (2004)).⁶⁴ Prior work provides several explanations for the downward trend, including wage compression (Hoxby and Leigh, 2004) and increasing labor market opportunities for women. My results suggest a third explanation: worsening working conditions in teaching. In other words, the workers who select into teaching are increasingly those with relatively low-paying next-best options who are willing to accept the pay premium teaching offers as a compensating differential. While the theory that relative job attributes may affect selection into teaching has received considerable attention (see e.g., Hanushek and Rivkin (2006)), this paper is one of the first to provide empirical evidence that supports this theory.

The second is that schools hold significant monopsony power over teachers (e.g., Ransom and Sims (2010)). Numerous studies document that, within the public school system, school districts hold monopsony power over teachers: for example, Ransom and Sims (2010) estimate that school districts in Missouri have the power to reduce wages by 27% compared to a frictionless world in which workers are informed and mobile. The authors posit that school districts hold market power because teachers have strong locational preferences (e.g., Boyd et al. (2013)), and because switching districts after one has begun accruing in a schools' pension plan can induce a significant financial loss. However, less attention has been paid to the idea that schools may hold market power in the broader labor market thanks to teaching offering uncommon yet desirable amenities, like job security and "summers off." My findings that teachers highly value such school-specific amenities suggest that schools may be able to reduce wages *and* sustain relatively challenging working conditions due to the combination of their market power across districts and in the broader labor market.

The third is that teachers in the US earn less than their similarly-educated counterparts. Numerous high-profile policy reports use the Current Population Survey (CPS) to show that teachers earn less than other college graduates (Allegretto and Mishel, 2018), and use these estimates of the observational pay gap to argue that teachers are undercompensated relative to their alternative job options. However, the results of other studies bring into question whether

⁶⁴The findings of this literature raise a natural question: is academic performance in college, including on standardized tests such as the Praxis, related to teacher effectiveness? This question is outside the scope of this paper. However, prior work examining this question generally finds that academic performance is not highly predictive of performance, as in e.g., Harris and Sass (2011).

the observational pay gap is an unbiased estimate of the true pay gap between teaching and teachers’ next-best options. Event studies examining how pay changes for exiting teachers in the National Longitudinal Survey of Youth (NLSY) (Stinebrickner, 2002), Georgia (Scafidi, Sjoquist, and Stinebrickner, 2006), and Washington (Goldhaber et al., 2022) consistently find that exiting teachers tend to leave to lower-paying jobs. My findings show that these two seemingly opposed results are not at odds with one another: I find that while teachers are indeed paid less than other college-educated workers, their next-best job options actually pay less than teaching, consistent with evidence from event studies in other states. However, I also show that this large pay difference does not imply that teachers are “overcompensated” overall: instead, my findings suggest that teaching is a high-paying option relative to teachers’ next-best options *because* it offers a compensating differential.

Limitations. A notable limitation of this study is that the main results come from one state, which limits the generalizability of the findings. The pay and working conditions gaps likely vary by state, as suggested by the fact that the event study estimates from the matched CPS data—which covers the entire U.S., as opposed to one state—give a smaller, but still positive, pay gap between teaching and teachers’ next-best jobs. That said, the few of existing studies that examine whether workers who exit teaching leave to higher-or lower-paying jobs consistently find that exiting teachers leave to lower-paying jobs, suggesting that the pay premium I find is not unique to KY.

V. CONCLUSION

This paper revisits the long-standing belief that teachers, as public sector workers who are largely unionized, are able to extract rents. This belief is at odds with recent reports of teacher shortages and challenging working conditions in teaching. To reconcile these conflicting reports, I investigate how the *overall* compensation in teaching today compares to teachers’ next-best jobs, taking both pay and working conditions into account.

I present a new approach for evaluating an occupation-specific rent and apply it to teaching, which entails estimating and comparing the gap in pay to the gap in amenity value between teaching and teachers’ next-best jobs. I use two quasi-experimental strategies to estimate the pay gap. I develop an empirical strategy to identify the treatment effects of becoming a teacher using a regression kink design, leveraging the fact that test-takers can retake the exam if they do not pass on their first attempt. I also use an event study design to estimate the effect of

leaving teaching on earnings. I estimate both models using novel linked data that allow me to track the short-term earnings, education, and employment outcomes for individuals who express an interest in pursuing teaching by taking the teaching entry exam in KY. The data and strategies allow me to provide the first quasi-experimental evidence characterizing teachers' next-best options, in terms of pay and occupation, in the US.

To estimate the gap in amenity values, I field a survey to current and former teachers in KY to elicit information about their working conditions that includes a series of stated preference experiments. My survey instrument is identical on key dimensions to the instrument used in the American Working Conditions Survey (AWCS) by [Maestas et al. \(2023\)](#), allowing me to compare the jobs of KY teachers to other jobs in the U.S. captured by the AWCS. My survey is the first to systematically characterize the working conditions in teaching in a way that allows for direct comparison with teachers' next-best options.

From the RK and event study, I find that teaching offers a pay premium: interested teachers have substantially lower-paying next-best options than teaching. From the survey, I find that teaching offers less valuable working conditions than teachers' next-best jobs, particularly for inexperienced teachers. Ultimately, the combined estimates suggest that experienced teachers earn a moderate rent of 16% of their salary, while inexperienced teachers earn no rent at all. These estimates, which come from two independent designs and datasets, indicate that the large pay gap that exists is primarily a compensating differential rather than a rent.

The idea that disamenities may account for a sizeable share of the labor cost of teaching suggests that there could be large benefits in evaluating the costs and benefits of improving workplace amenities for teachers. Which working conditions does teaching lack that can be introduced or improved at low cost? And what is the disamenity value associated with current policy proposals that aim to improve student outcomes but worsen teachers' working conditions, like standardized testing or banned disciplinary policies? Investigating these questions may be a critical step in addressing the hiring challenges that schools are facing today.

MICROSOFT RESEARCH

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VI. TABLES

Table I: Summary Statistics on First-Time Test-Takers

	Mean	S.D.	<i>N</i>
<i>Panel A: Sample characteristics at first attempt</i>			
Age	24.35	7.57	10,127
Female	0.76	0.43	10,127
White	0.92	0.28	10,127
First-time enrollee in post-secondary institution	0.77	0.42	10,127
Holds a Pell grant	0.20	0.40	10,127
Earnings 1yr prior (2018 USD)	11,384.91	12,917.88	6,744
<i>Panel B: Praxis performance and retaking</i>			
Math score (passing cutoff: 150)	158.03	23.50	10,127
Pass math on first attempt	0.64	0.48	10,127
Among those who fail on first attempt...			
% who retake at least once	0.80		3,645
% who pass on first retake	0.33		3,645
Total number of attempts at the Praxis	1.63	1.39	10,127
<i>Panel C: Leakage from the teacher pipeline</i>			
Attempt Praxis for first time	1.00		10,127
Ever pass entire Praxis	0.80		10,127
Enroll in EPP within 3 years	0.72		10,127
Get certified within 4 years	0.53		10,127
Become teacher within 4 years	0.46		10,127

Note: Summary statistics on Praxis first-attempters between 2013 and 2017.

Table II: Estimates of the Effects of Passing the Praxis, Becoming Certified, and Becoming a Teacher on Employment and Earnings

	Outcome:		Outcome:	
	Pr(Employed in 4yrs)		Earnings in 4yrs, cond. on emp.	
	First Stage	Struct. Model	First Stage	Struct. Model
	(1)	(2)	(3)	(4)
<i>Treatment A: Passing the entire Praxis</i>				
RD estimate	0.12	-0.25		
(conventional std. error)	(0.02)	(0.23)		
VtF confidence interval		[-.7, .2]		
<i>Treatment B: Becoming Certified</i>				
RK estimate			-0.010	21,685.2
(conventional std. error)			(0.004)	(11,380.9)
VtF confidence interval				[-621.4, 43,991.8]
<i>Treatment C: Becoming a Teacher</i>				
RK estimate			-0.012	18,329.5
(conventional std. error)			(0.004)	(8232.3)
VtF confidence interval				[2194.2, 34,464.7]
Bandwidth	14	14	14	14
<i>N</i>	4,352	4,352	2,958	2,958

Note: All estimates are reported for the full sample of first-time Praxis test-takers between 2013-2017, using a bandwidth of 14 (i.e. comparing individuals with scores 136, 138, 140, 142, 144, 146, 148 to those with scores 150, 152, 154, 156, 158, 160, 162, 164). Conventional standard errors are reported in round brackets, and VtF confidence intervals are reported in [Lee et al. \(2024\)](#).

Table III: Benchmarking the Pay Gap Estimates to Teacher Salaries

	Pay Gap / Salary		
	Salary	Gap = \$18,000	Gap = \$21,000
	(1)	(2)	(3)
Most relevant benchmark for RK sample:			
2yrs experience, Rank I	\$45,000	40%	47%
Most relevant benchmark for event study sample:			
Avg earnings before exit	\$55,000	33%	38%

Note: Column (1) shows salaries from the 2019 salary schedules for teachers, available online from the KY Department of Education. Column (2) shows the ratio of the lower bound estimate of the pay gap to the salary in Column (1). Column (3) shows the ratio of the upper bound estimate of the pay gap to the salary in Column (1).

Table IV: Summary Statistics on Teachers in Survey versus Administrative data

	Survey	Administrative data
	(1)	(2)
<i>Panel A: Demographics</i>		
White	0.90 (0.30)	0.95 (0.21)
Woman	0.76 (0.43)	0.78 (0.41)
Resides in rural area	0.57 (0.49)	0.27 (0.45)
Age: 26-29	0.07 (0.25)	0.11 (0.31)
Age: 30-34	0.08 (0.27)	0.15 (0.35)
Age: 35-39	0.14 (0.34)	0.16 (0.37)
Age: 40-44	0.17 (0.38)	0.15 (0.36)
Age: 45-49	0.16 (0.37)	0.15 (0.36)
Age: 50 plus	0.33 (0.47)	0.23 (0.42)
Experience: 3-5 years	0.08 (0.28)	0.14 (0.35)
Experience: 5-10 years	0.14 (0.35)	0.19 (0.40)
Experience: 10+ years	0.72 (0.45)	0.57 (0.50)
Teaches at high school	0.35 (0.48)	0.26 (0.44)
Teaches at elementary school	0.63 (0.48)	0.48 (0.50)
Teaches at Title I school	0.68 (0.47)	0.67 (0.47)
<i>N</i>	1282	39385

Note: This table presents means (standard deviations in parentheses) describing teachers who responded to the survey in KY and the full KY teacher population in 2019.

VII. FIGURES

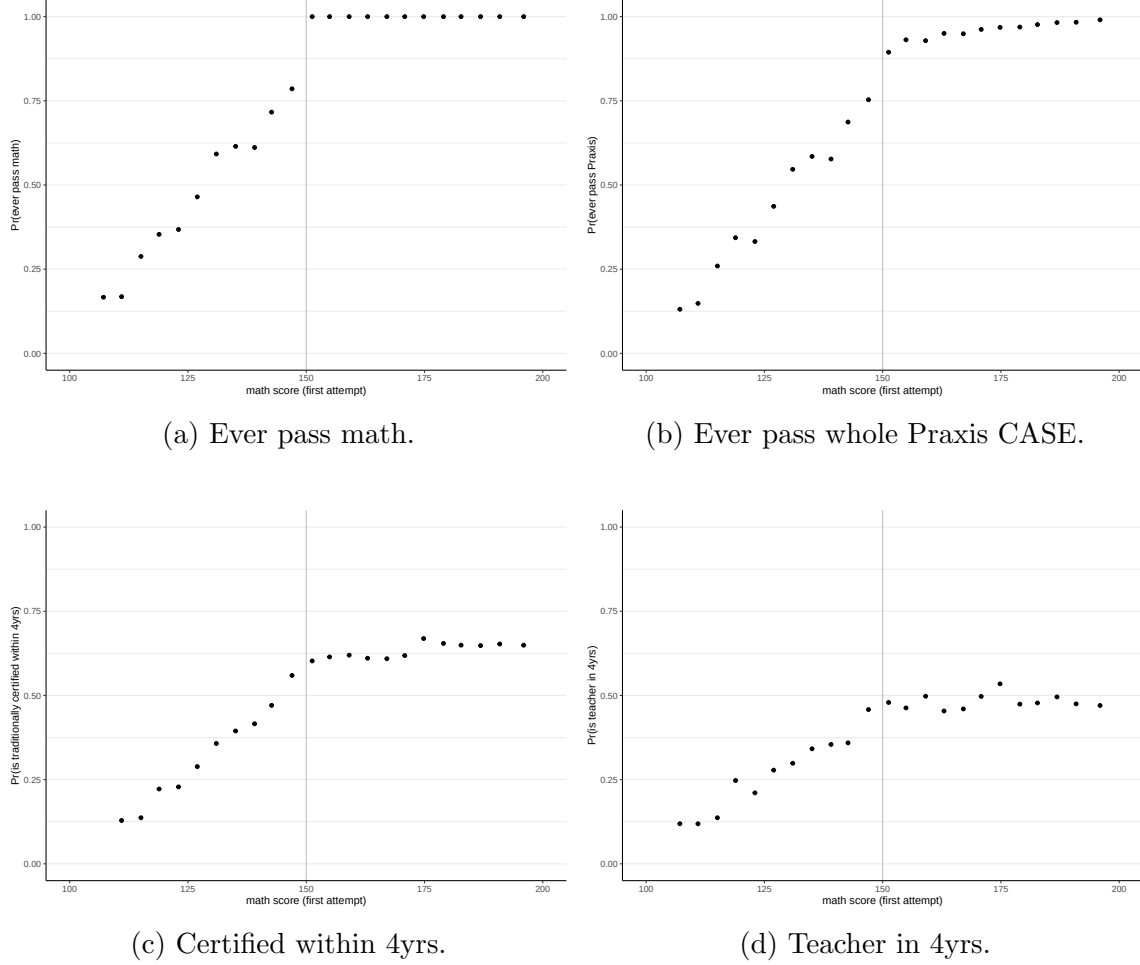
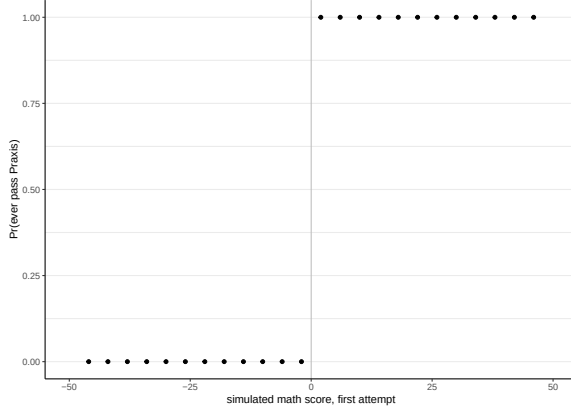
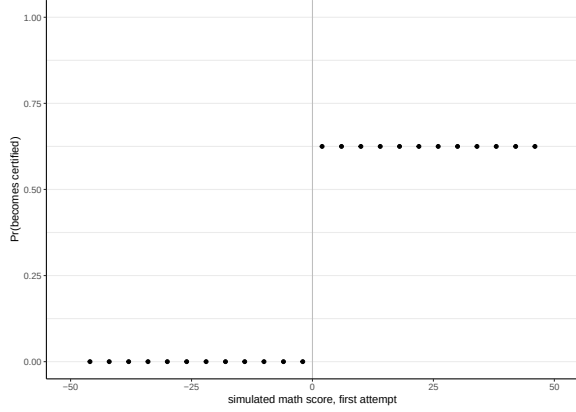


Figure I: First Stage Relationships between First-Attempt Scores and Certification Outcomes

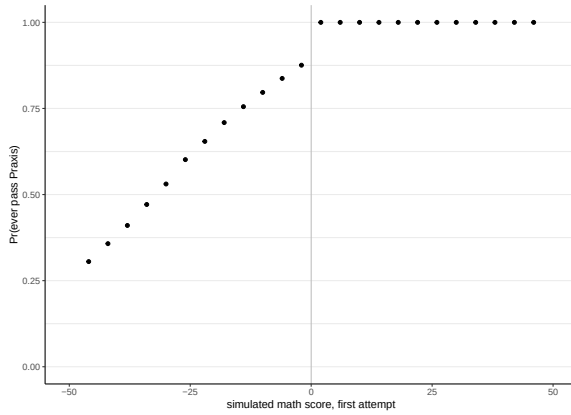
Note: Each panel shows the first stage relationship between the running variable (the first attempt math score) and the likelihood that individuals reach various stages along the teacher pipeline. Panel (a) shows the share of individuals who pass the math portion of the Praxis CASE by 2022, plotted over bins of the first-attempt math scores. Similarly, Panel (b) shows the share of individuals who pass all of math, reading, and writing on the Praxis CASE by 2022; Panel (c) shows the share of individuals who obtain their teacher certification within 4 years of their first attempt, and Panel (d) shows the share of individuals who are observed working as a teacher in KY public schools in the fourth year after their first attempt.



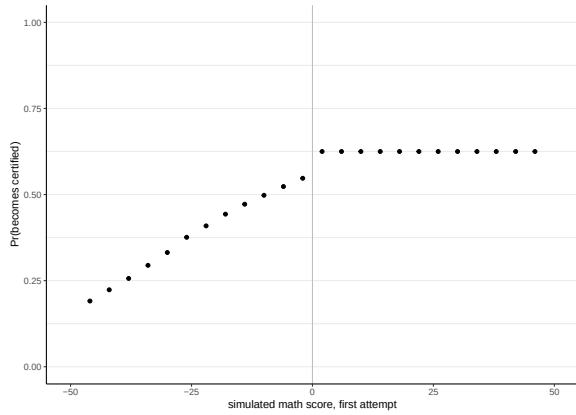
(a) Simulated $\Pr(\text{pass})$, one-shot



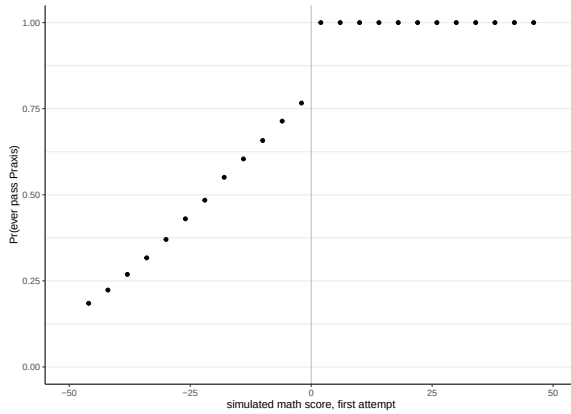
(b) Simulated $\Pr(\text{certified})$, one-shot



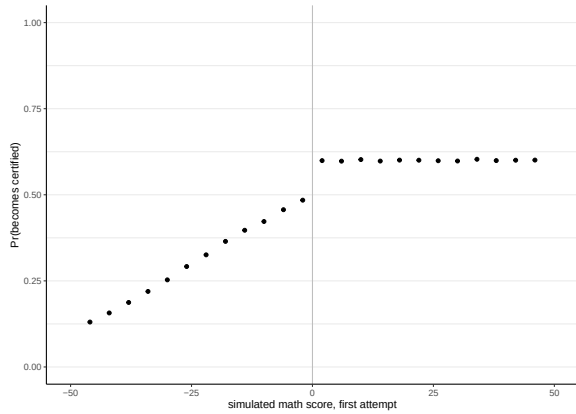
(c) Simulated $\Pr(\text{pass})$, with automatic retaking



(d) Simulated $\Pr(\text{certified})$, with automatic retaking



(e) Simulated $\Pr(\text{pass})$, with behavioral retaking



(f) Simulated $\Pr(\text{certified})$, with behavioral retaking

Figure II: Progression of first-stages from simulated one-shot test (1st row), to simulated dynamic test-taking with automatic retaking (2nd row), to simulated dynamic test-taking with behavioral retaking (3rd row).

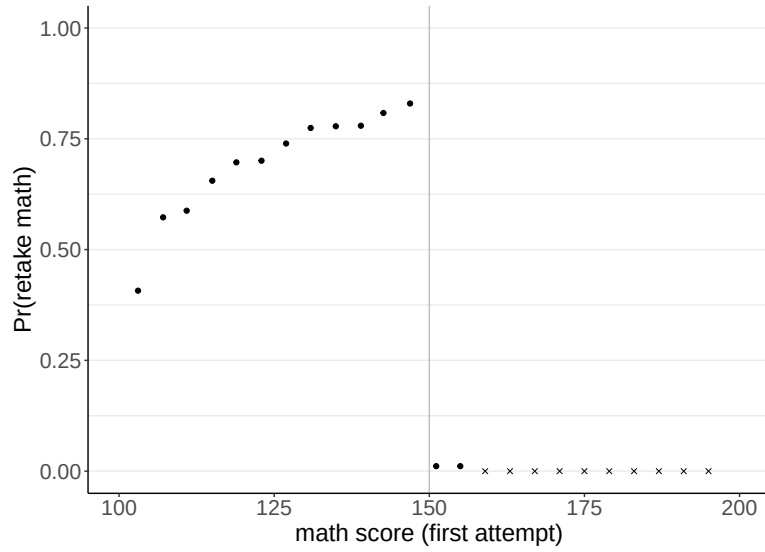
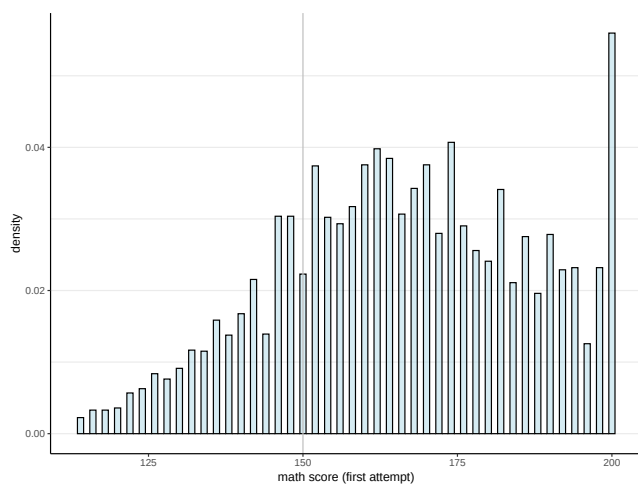
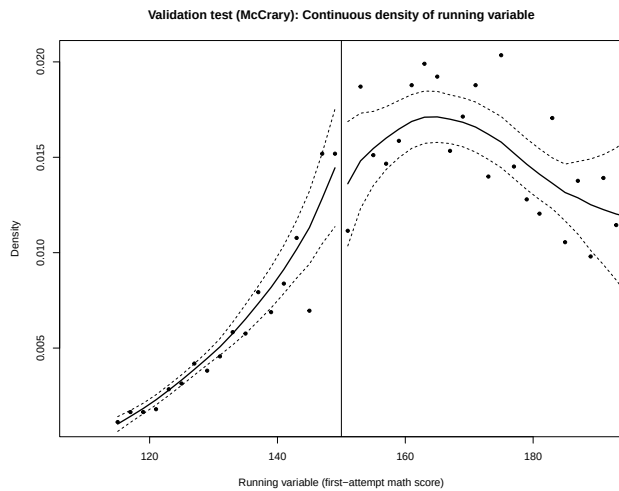


Figure III: Retaking Behavior

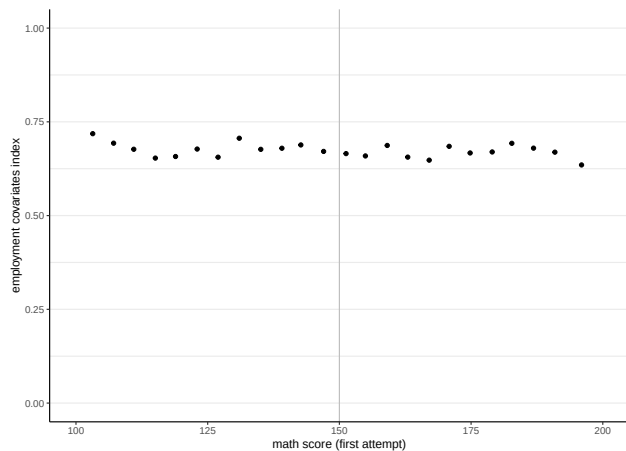
Note: This figure is a binscatter of the share of individuals who retake the exam at least once, plotted over first-attempt math scores. X's represent cells with too few observations to show. The increasing rate of retaking approaching the cutoff from the left, and the discontinuity in retaking after the cutoff, drives the kink.



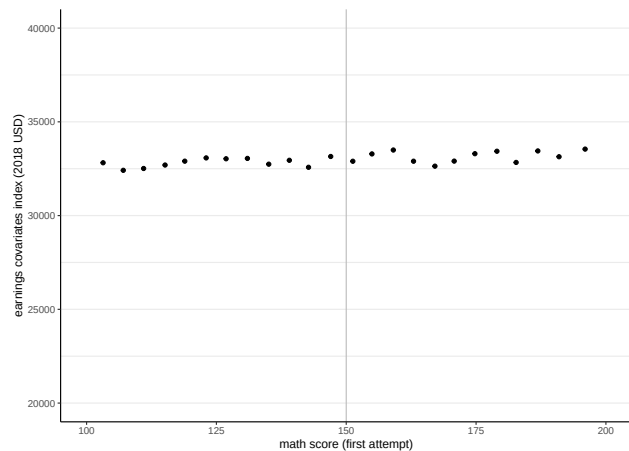
(a) Distribution of first-attempt math scores



(b) Manipulation test



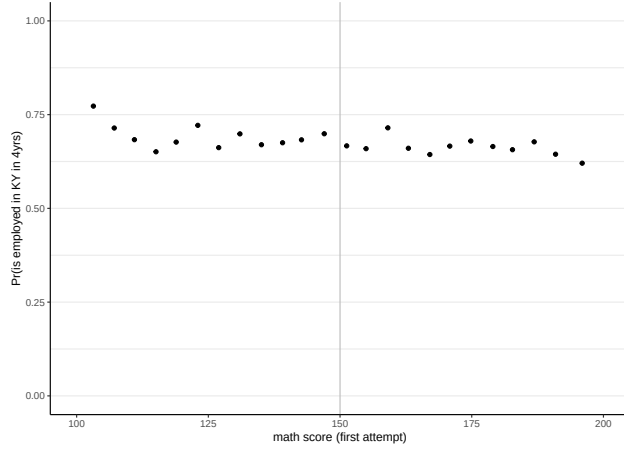
(c) Predicted $\Pr(\text{employed in 4yrs})$



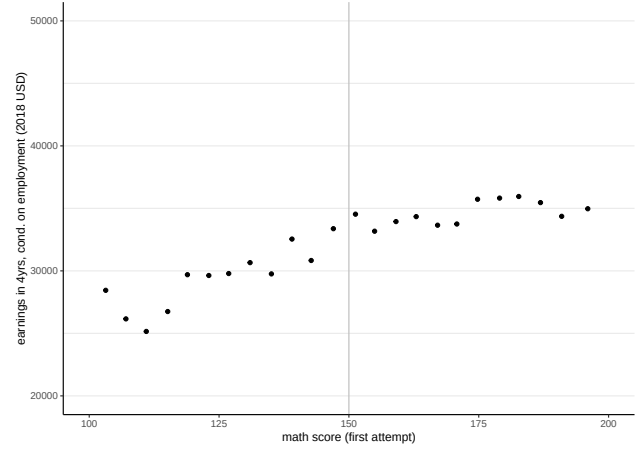
(d) Predicted earnings in 4yrs

Figure IV: Tests for Manipulation of First-Attempt Math Scores AND Tests of Selection Around the Cutoff, Using Covariates Index

Note: Panel (a) shows a histogram of first-attempt math scores for the sample that passed reading and writing on the first attempt. Panel (b) conducts the test in [McCrary \(2008\)](#) and shows fitted polynomials to the right and left of the test score cutoff on the same sample used to plot the histogram in Panel (a). AND Each panel presents binscatters for an “index” variable that is predicted using only covariates at “baseline,” i.e. in the year of or one year before attempting the Praxis for the first time. The index plotted in Panel (a) is the predicted share of individuals who are employed four years after the test based on the following covariates: gender, race/ethnicity, age, a dummy for whether the person was enrolled in post-secondary at the time of the test, a dummy for whether the person was enrolled in public post-secondary at the time of the test, a dummy for whether the person held a Pell grant at the time of the test, a dummy for whether the individual already possessed a post-secondary degree at the time of the test, earnings one year before the first attempt, a dummy for whether the person was employed one year before the attempt, dummies for the industry the person worked in one year before the attempt, and the year and month of the test. The index plotted in Panel (b) is the predicted earnings four years after the first attempt using the same set of baseline covariates.



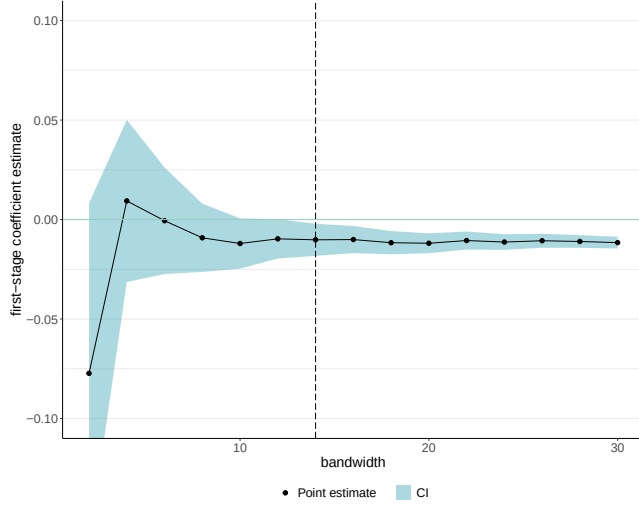
(a) $\Pr(\text{employed in 4yrs in KY})$.



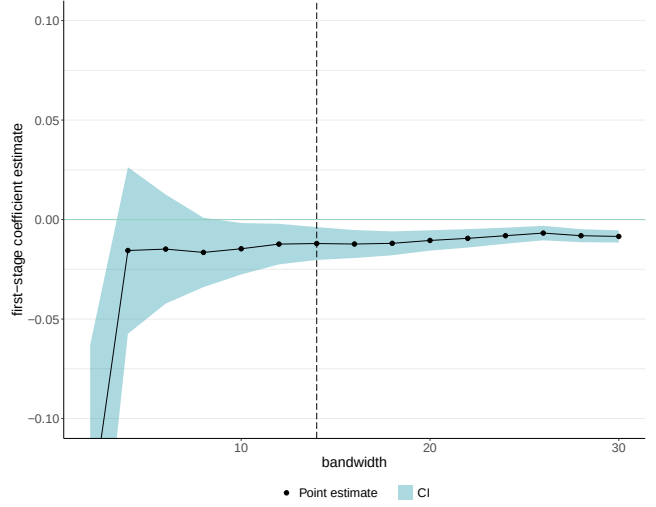
(b) Earnings in 4yrs (2018 USD).

Figure V: Reduced Form Relationships underlying Regression Kink Design

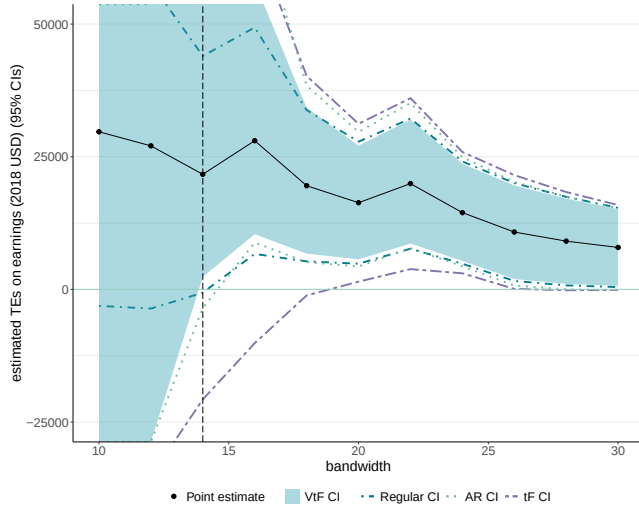
Note: Binscatters of the likelihood of being employed and earnings 4 years after one's first attempt on the Praxis CASE math, plotted over bins of first-attempt math scores.



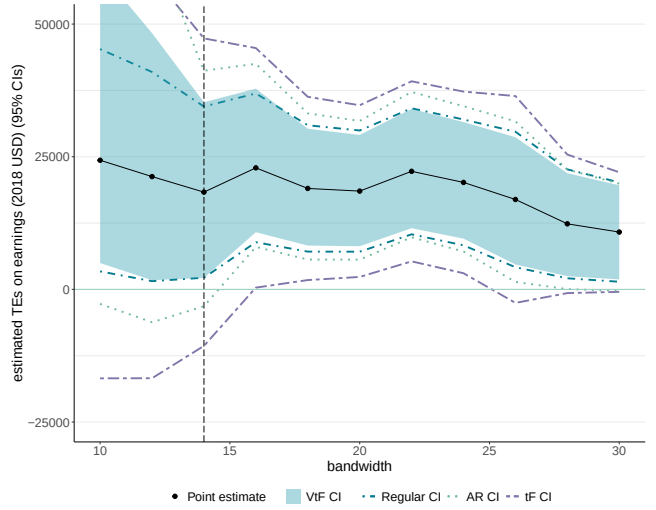
(a) First stage: Becoming certified.



(b) First stage: Becoming a teacher.



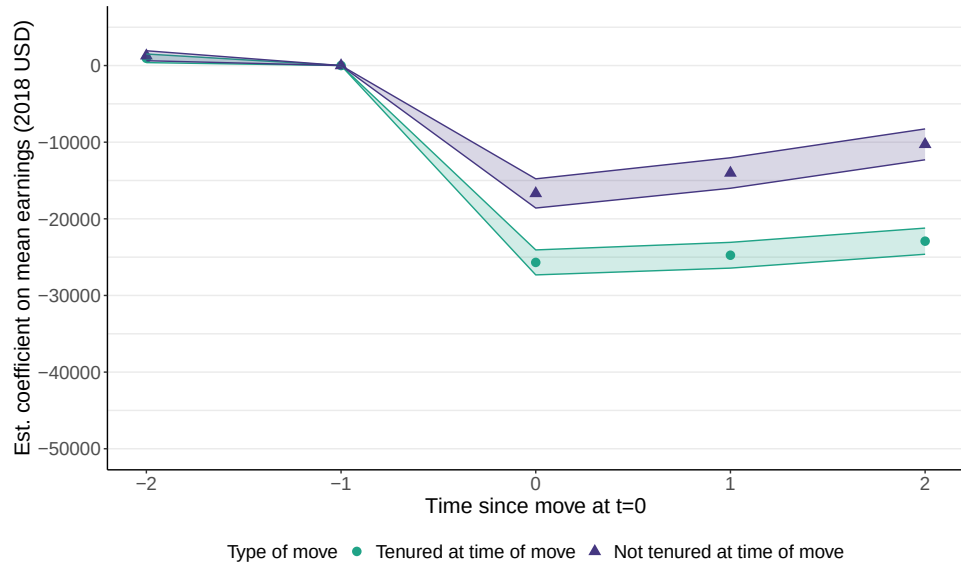
(c) Effect of becoming certified on earnings.



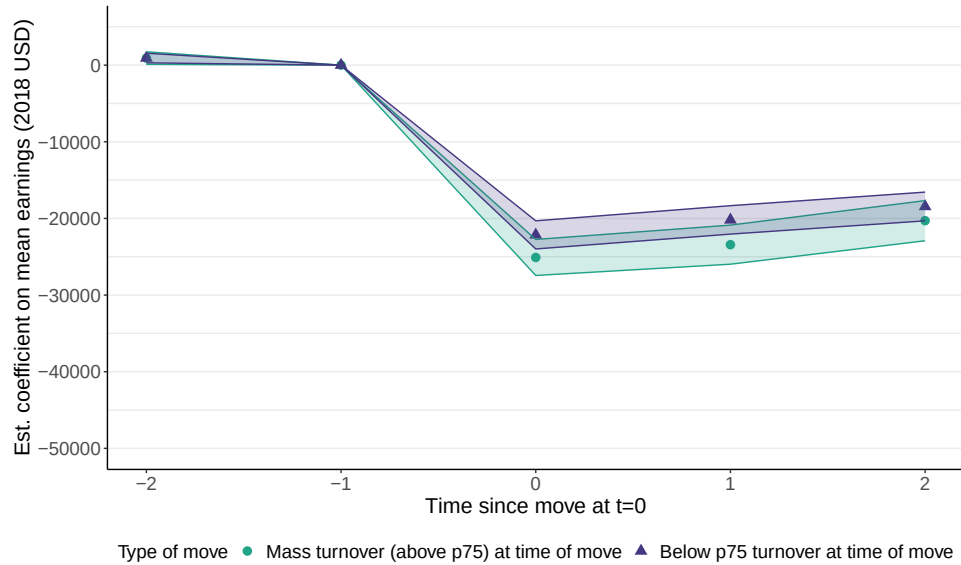
(d) Effect of becoming a teacher on earnings.

Figure VI: Estimates from the Regression Kink Design

Note: The first row shows estimates from first stage regressions over bandwidths ranging from 2 to 30. The treatment variable in Panel (a) is the likelihood of becoming certified in 4 years, meaning the coefficients can be interpreted as the change in the slope of $\Pr(\text{certified})$ at the cutoff. The treatment variable in Panel (b) is the likelihood of becoming a teacher in 4 years after the first attempt. The second row shows the 2SLS coefficients over bandwidths from 10–30; smaller bandwidths are omitted because the first stage estimates become stably significant starting at a bandwidth of 10. Panel (c) plots the 2SLS estimates of the effect of becoming certified on earnings, while Panel (d) plots the 2SLS estimates of the effect of becoming a teacher on earnings, 4 years after taking the Praxis for the first time. Multiple 95% confidence intervals (CIs) are shown following recommended standard error corrections for weak instruments, including: Anderson-Rubin CIs, tF CIs (Lee et al., 2022), and VtF CIs (Lee et al., 2024).



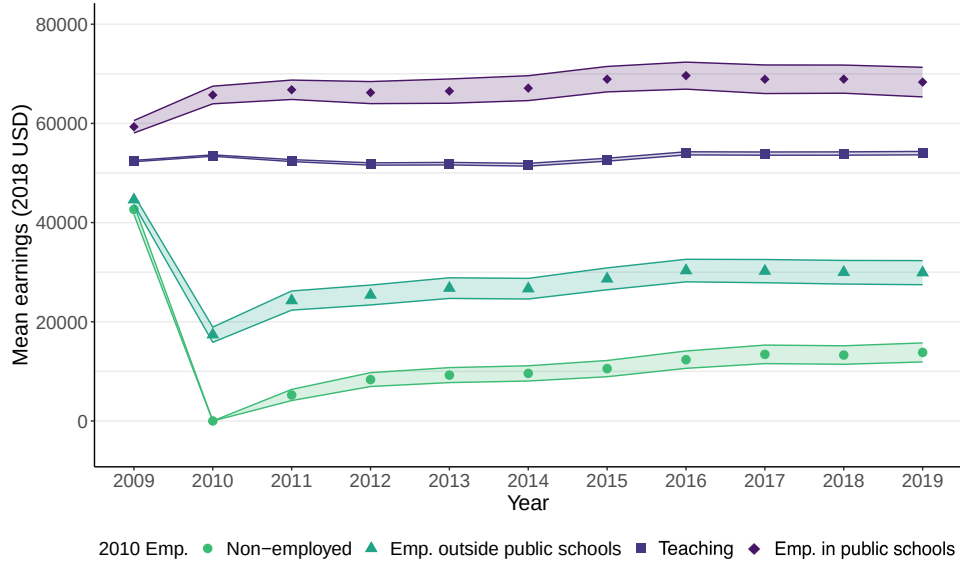
(a) Est. earnings differences relative to $t = -1$, by movers' tenure status



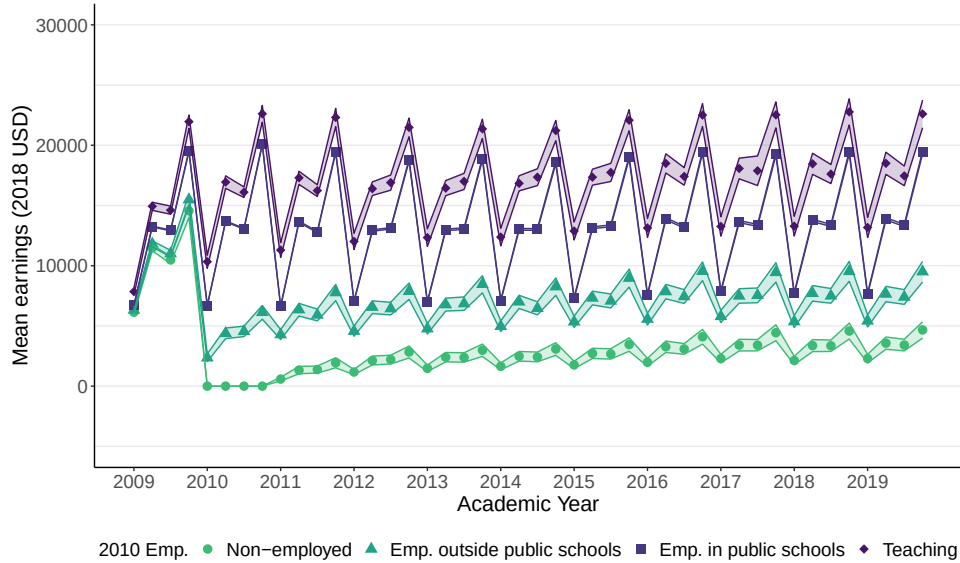
(b) Est. earnings differences relative to $t = -1$, by degree of turnover at school

Figure VII: Event Study Estimates of Earnings Changes around Teacher Exits

Note: Event study estimates for the sample of teachers who exit the public school system and maintain at least 3 quarters of wages in the following years. The outcome is total earnings over the year. Observation counts are the same for every point shown. Non-employment is coded as 0.



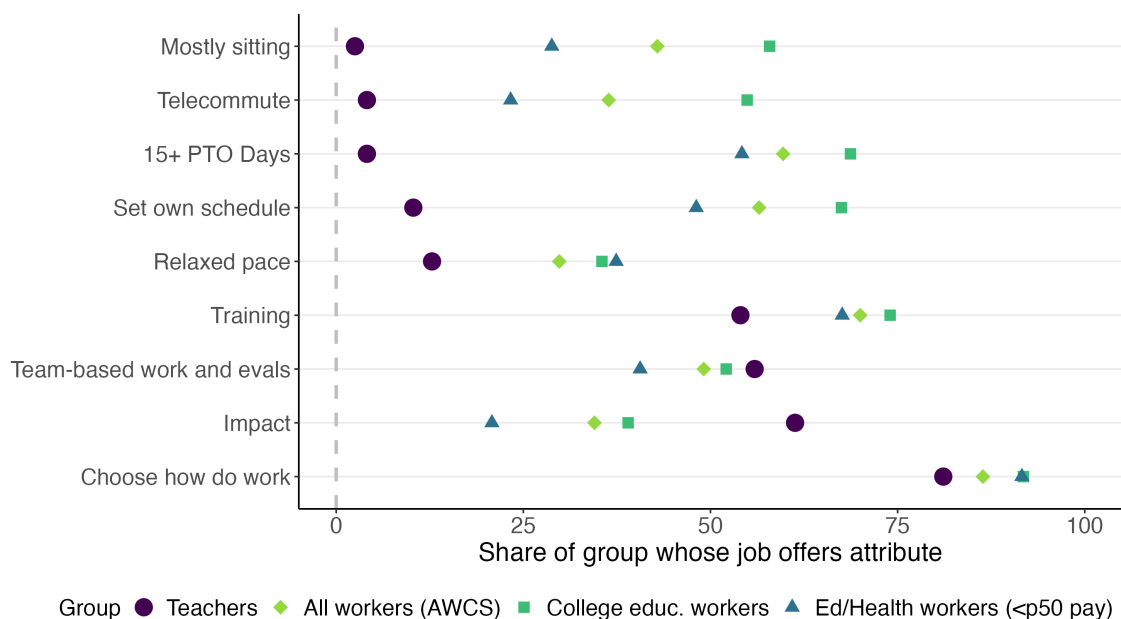
(a) Annual earnings.



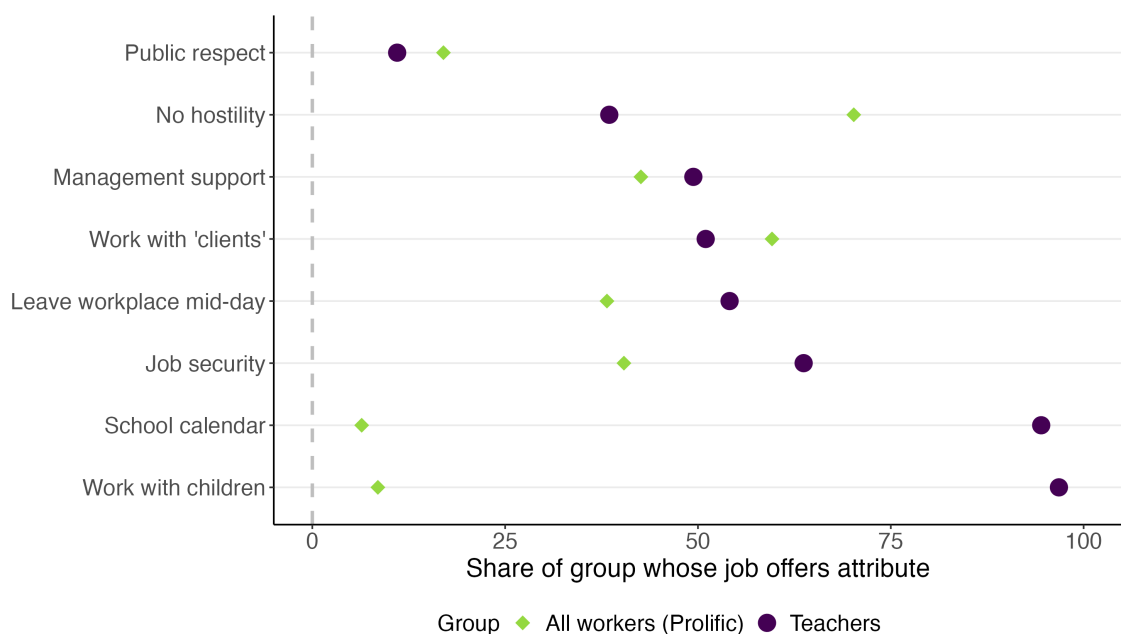
(b) Quarterly earnings.

Figure VIII: Exiting Teachers' Earnings Trajectories

Note: Each figure plots earnings trajectories for teachers in 2009, grouped by the type of transition they made in 2010. The four transition groups are: teachers who stay teachers in 2010 (squares), teachers who exit teaching but continue being employed in public schools in 2010 (diamonds), teachers who exit teaching but continue being employed in 2010 (triangles), and teachers who exit teaching and do not appear in the UI data in 2010 (circles). Earnings for those not in the UI data are coded as 0. Panel (a) shows annual earnings, constructed by aggregating quarterly earnings, which are shown in Panel (b). An academic year is defined as the first two quarters of the year and the last two quarters from the previous year.



(a) AWCS attributes



(b) Teaching-specific attributes

Figure IX: The Incidence of Working Conditions Across Jobs

Note: This figure shows the percentage of workers who report having each of 17 desirable attributes. Figure (a) shows 4 groups: KY teachers (circles, from my survey), all workers in the US (diamonds, from the AWCS), college educated workers in the US (squares, from the AWCS), and workers in the education and health sector with below-median earnings in their sectors (triangles, from the AWCS). Figure (b) shows 2 groups: KY teachers (circles, from my survey) and all workers (diamonds, from my survey). The attributes are sorted from least common in teaching (top) to most common in teaching (bottom).

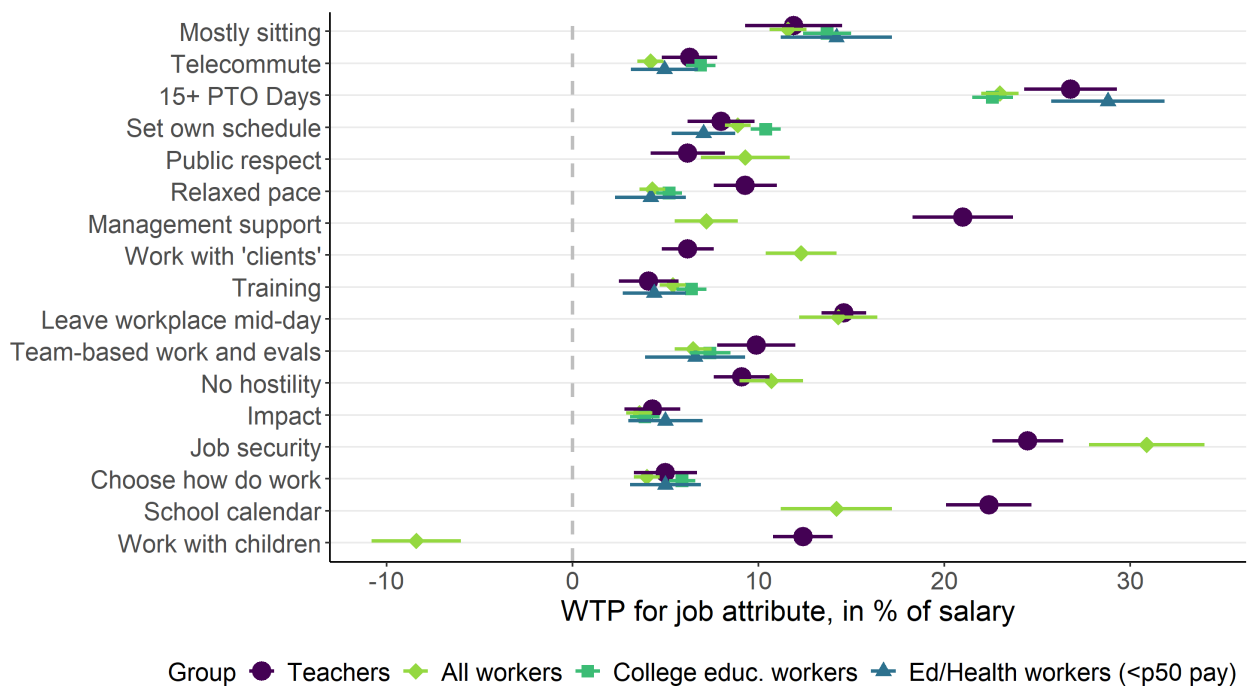


Figure X: Willingness-to-pay Estimates

Note: This figure shows willingness-to-pay estimates for each of the nine job attributes and for 4 different groups: KY teachers (circles, from my survey), all workers in the US (diamonds, from the AWCS and my survey), college-educated workers (squares, from the AWCS), and workers in the education and health sector with below-median earnings in their sectors (triangles, from the AWCS). Note that for the attributes originally from the AWCS, the “all workers” estimates use the AWCS estimates, while for the teaching-specific attributes from my survey, the “all workers” estimates use my survey data. The attributes are sorted from least common in teaching (top) to most common in teaching (bottom).

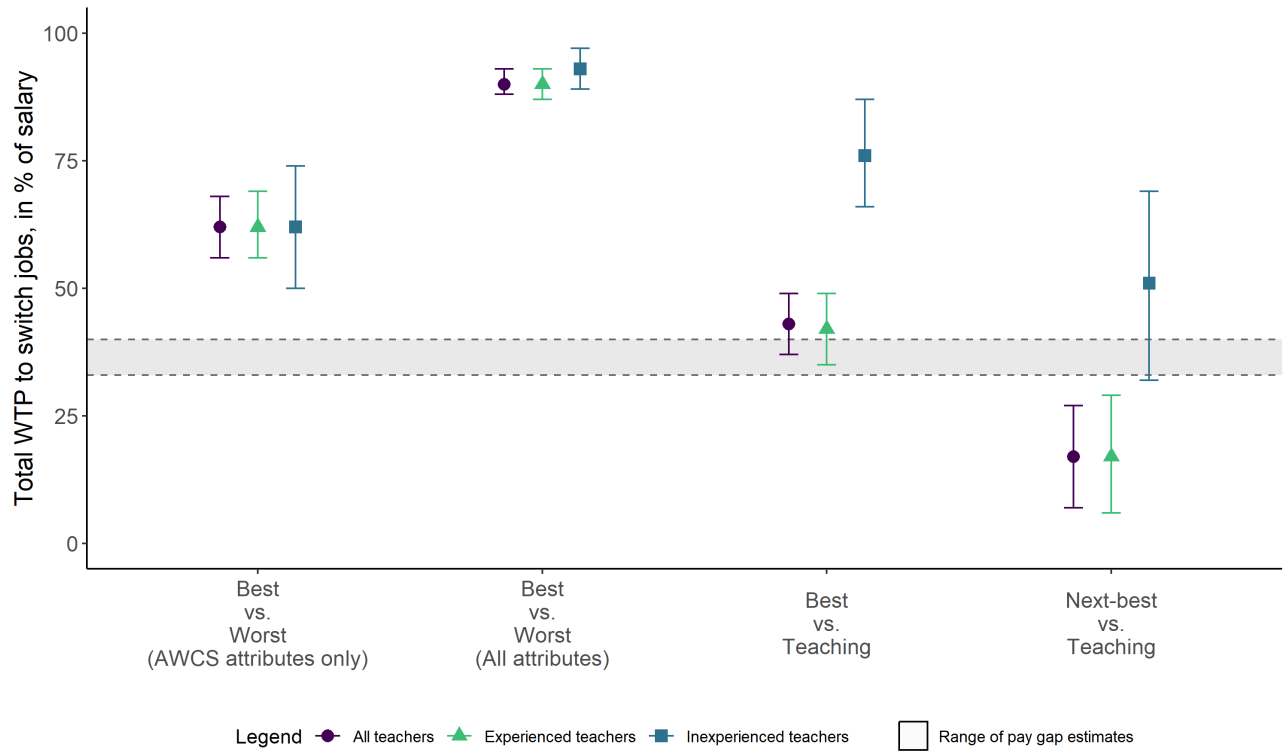


Figure XI: Estimates of Total Willingness-to-pay to Switch Jobs, By Experience

Note: This figure shows estimates of KY teachers' willingness to pay for one bundle of attributes over another, using data from the choice experiments. 95% confidence intervals are shown, with standard errors estimated using the delta method and clustered at the respondent level. The shaded region shows the range of pay gap estimates from the quasi-experimental designs in the first half of the paper, ranging from 33-40% of the teachers' salary.



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SUPPLEMENTARY MATERIAL

I.A. Identification Proofs

Here I provide proofs showing that under certain assumptions, the RD and RK estimands deliver weighted treatment effects of interest in settings where discontinuities arise (due to e.g. test score cutoffs) and retaking is allowed.

Notation. For exposition, assume that individuals can only retake the exam once. We use the following notation throughout the proofs that follow.

- U : person type (unobserved heterogeneity)
- S_{1i} : first-attempt score, drawn from $f_{S_1|U=u}(s)$
- R : indicator for whether one decides to retake the exam
- S_{2i} : second-attempt score, drawn from $f_{S_2|U=u,R=1}(s)$.
- $p_r(S_1, U)$: the probability that a person of type U retakes the test
- $p_c(U)$: the probability that a person of type U becomes certified, conditional on having passed the test at any point (first or second attempt)
- $y(C, U)$: potential earnings outcomes, where $Y_{C,i} = y(1, U)$ and $Y_{NC,i} = y(0, U)$.

RD Design. Suppose there were a discontinuity in the observed likelihood of individuals becoming certified at the cutoff, resulting from the random assignment of individuals to the left or right of the cutoff. Then the 2SLS estimand that results from using whether the first-attempt score is on the right or left of the cutoff as an instrument for whether one becomes certified to teach is

$$\frac{\lim_{s \rightarrow 0^+} \mathbb{E}[y(C, U) \mid S_1 = s] - \lim_{s \rightarrow 0^-} \mathbb{E}[y(C, U) \mid S_1 = s]}{\lim_{s \rightarrow 0^+} \mathbb{E}[C \mid S_1 = s] - \lim_{s \rightarrow 0^-} \mathbb{E}[C \mid S_1 = s]}. \quad (10)$$

The following proof derives expressions for each of the four terms in (10) and shows that, when pieced together, they deliver a weighted average treatment effect.

Beginning with the numerator, observe that $\mathbb{E}[y(C, U) \mid S_1 = s]$ can be written as

$$\begin{aligned} \mathbb{E}[y(C, U) \mid S_1 = s] &= \int \mathbb{I}\{s \geq 0\} [p_c(u)y(1, u) + (1 - p_c(u))y(0, u)] \\ &\quad + \mathbb{I}\{s < 0\}(1 - p_r(s, u))y(0, u) \\ &\quad + \mathbb{I}\{s < 0\}p_r(s, u)\Pr(S_2 < 0 \mid u, R = 1)y(0, u) \\ &\quad + \mathbb{I}\{s < 0\}p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1) [p_c(u)y(1, u) + (1 - p_c(u))y(0, u)] dF_{U|S_1=s}(u). \end{aligned} \tag{11}$$

In words, (11) says that there are four possible routes one can take to reach a potential earnings outcome. First, they could pass the Praxis on their first attempt and either get certified or not. Second, they could fail the Praxis on their first attempt and choose not to retake the exam. Third, they could fail on their first attempt, choose to retake the exam, but fail again. Finally, they could fail on their first attempt, choose to retake, and pass. Each route leads either to the potential earnings outcome $y(1, U)$ or $y(0, U)$.

Breaking the expression up in this fashion makes it clear which terms are relevant when evaluating limits from the right and left of the cutoff. When taking the limit from the right of $s = 0$, we can ignore the three routes that involve $s < 0$ and only focus on the route that involves $s \geq 0$:

$$\begin{aligned} \lim_{s \rightarrow 0^+} \mathbb{E}[y(C, U) \mid S_1 = s] &= \lim_{s \rightarrow 0^+} \int p_c(u)y(1, u) + (1 - p_c(u))y(0, u) dF_{U|S_1=s}(u) \\ &= \lim_{s \rightarrow 0^+} \int y(0, u) + p_c(u)(y(1, u) - y(0, u)) dF_{U|S_1=s}(u) \\ &= \lim_{s \rightarrow 0^+} \int y(0, u) + p_c(u)(y(1, u) - y(0, u)) \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u), \end{aligned} \tag{12}$$

where the substitution in the last line follows from Bayes rule.

Similarly, taking the limit from the left yields

$$\begin{aligned}
\lim_{s \rightarrow 0^-} \mathbb{E}[y(C, U) \mid S_1 = s] &= \lim_{s \rightarrow 0^-} \int (1 - p_r(s, u))y(0, u) \\
&\quad + p_r(s, u)\Pr(S_2 < 0 \mid u, R = 1)y(0, u) \\
&\quad + p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1) [p_c(u)y(1, u) + (1 - p_c(u))y(0, u)] dF_{U|S_1=s}(u) \\
&= \lim_{s \rightarrow 0^-} \int y(0, u) \\
&\quad + p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1)p_c(u) (y(1, u) - y(0, u)) \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u).
\end{aligned} \tag{13}$$

Then under the assumption that $f_{S_1|u}(s)$ and $f_{S_1}(s)$ are everywhere continuous and smooth through the cutoff $s = 0$, the numerator of the RD estimand is

$$\begin{aligned}
&\lim_{s \rightarrow 0^+} \mathbb{E}[y(C, U) \mid S_1 = s] - \lim_{s \rightarrow 0^-} \mathbb{E}[y(C, U) \mid S_1 = s] \\
&= \int (y(1, u) - y(0, u)) \left[\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u)p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1) \right] \frac{f_{S_1|u}(0)}{f_{S_1}(0)} dF_U(u).
\end{aligned} \tag{14}$$

We can expand the denominator of (10) in a similar fashion. First, since C is a binary treatment, observe that we can write

$$\begin{aligned}
\Pr[C = 1 \mid S_1 = s] &= \int \mathbb{I}\{s \geq 0\}p_c(u) \\
&\quad + \mathbb{I}\{s < 0\}p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1)p_c(u) dF_{U|S_1=s}(u).
\end{aligned} \tag{15}$$

Taking limits from the right and left of $s = 0$ yields

$$\begin{aligned}
\lim_{s \rightarrow 0^+} \Pr[C = 1 \mid S_1 = s] &= \lim_{s \rightarrow 0^+} \int p_c(u) \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u), \text{ and} \\
\lim_{s \rightarrow 0^-} \Pr[C = 1 \mid S_1 = s] &= \lim_{s \rightarrow 0^-} \int p_c(u)p_r(s, u)\Pr(S_2 \geq 0 \mid u, R = 1) \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u).
\end{aligned} \tag{16}$$

Again under the assumption that $f_{S_1|u}$ and f_{S_1} are smooth around the cutoff $s = 0$, the

denominator of the RD estimand is given by

$$\begin{aligned} & \lim_{s \rightarrow 0^+} \Pr[C = 1 \mid S_1 = s] - \lim_{s \rightarrow 0^-} \Pr[C = 1 \mid S_1 = s] \\ &= \int \left[\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) p_r(s, u) \Pr(S_2 \geq 0 \mid u, R = 1) \right] \frac{f_{S_1|u}(0)}{f_{S_1}(0)} dF_U(u). \end{aligned} \quad (17)$$

Equation (17) has an important interpretation: it gives the size of the discontinuity in the likelihood that one becomes certified. In other words, it is the effect of barely passing the Praxis on the first-attempt (versus barely failing) on the likelihood of becoming certified. This expression demonstrates that identification relies on there being a non-zero discontinuity; otherwise, the denominator would be zero and the estimand would be undefined.

From (17), we also see that depending on the setting, the discontinuity can be positive or negative. Although $p_c(u)$ is smooth through the cutoff, the retaking function $p_r(s, u)$ is not; those randomly assigned to the right of the cutoff retake with probability zero, while those randomly assigned to the left may retake with a positive probability. The discontinuity could be positive if those to the left retake with some constant probability that is independent of u . But, the discontinuity could be negative if for those with $s < 0$, $p_r(s, u)$ is only positive for people with high $p_c(u)$, leading to

$$\lim_{s \rightarrow 0^-} p_c(u) p_r(s, u) \Pr(S_2 \geq 0 \mid u, R = 1) > \lim_{s \rightarrow 0^+} p_c(u).$$

Complete expression. Finally, we obtain an expression for the 2SLS estimand by dividing the numerator given by (14) by the denominator given by (17). The resulting expression is as previously described: a weighted average of the heterogeneous treatment effects $y(1, u) - y(0, u)$ over the distribution of u :

$$\frac{\lim_{s \rightarrow 0^+} \mathbb{E}[y(C, U) \mid S_1 = s] - \lim_{s \rightarrow 0^-} \mathbb{E}[y(C, U) \mid S_1 = s]}{\lim_{s \rightarrow 0^+} \mathbb{E}[C \mid S_1 = s] - \lim_{s \rightarrow 0^-} \mathbb{E}[C \mid S_1 = s]} = \int [y(1, u) - y(0, u)] \varphi_{RD}(u) dF_U(u), \quad (18)$$

where the weights are given by

$$\varphi_{RD}(u) = \frac{[\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) p_r(s, u) \Pr(S_2 \geq 0 \mid u, R = 1)] \frac{f_{S_1|u}(0)}{f_{S_1}(0)}}{\int [\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) p_r(s, u) \Pr(S_2 \geq 0 \mid u, R = 1)] \frac{f_{S_1|u}(0)}{f_{S_1}(0)} dF_U(u)}. \quad (19)$$

We can interpret the $\phi_{RD}(u)$ terms as weights as they are positive for all u . The RD estimand

is most heavily weighted towards “compliers” as defined in Supplementary Materials [I.C](#): individuals who either (1) pass on their first attempt and thus get certified, or (2) fail on their first attempt and do not get certified.⁶⁵

RK Design. The above section shows that if there were an observed discontinuity in the likelihood of certification, an RD design would identify a weighted average treatment effect.

However, in this paper’s setting, the random assignment of individuals to the right or left of the cutoff does not induce a discontinuity in the observed likelihood of certification, meaning the RD design cannot be applied. Instead, the same random assignment induces a kink (i.e., a change in the slope) in the likelihood of certification at the cutoff. The following proof demonstrates that in such an environment, under a few additional assumptions, an RK design can be used to identify a weighted average treatment effect of interest.

The 2SLS estimand that arises from using (whether one scores to the right or left of the cutoff) $\times$ (the distance of their score from the cutoff) as an instrument for getting certified is given by

$$\frac{\lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[y(C,U)|S_1=s]}{ds} - \lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[y(C,U)|S_1=s]}{ds}}{\lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[C|S_1=s]}{ds} - \lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[C|S_1=s]}{ds}}. \quad (20)$$

The following proof follows the same general steps as the “discontinuity” proof above: we derive expressions for each of the four terms that appears in (20) and show that pieced together they deliver another weighted average treatment effect.

Notation. For notational convenience, we will use $\Lambda(s)$ to denote the following expression which appears throughout the proof:

$$\Lambda(s) \equiv \frac{\partial \left(\frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right)}{\partial s} = \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \left[\frac{f'_{S_1|u}(s)}{f_{S_1|u}(s)} - \frac{f'_{S_1}(s)}{f_{S_1}(s)} \right]. \quad (21)$$

Numerator. We begin by expanding the numerator. We already derived the general expression for $\mathbb{E}[y(C,U) | S_1 = s]$ in (11). Taking derivatives and limits of (11) from the right

⁶⁵Note that, at a glance, the interaction between $p_c(u)$ and $(1 - \lim_{s \rightarrow 0^-} p_r(s, u) \Pr(S_2 \geq 0 | u, R = 1))$ makes it appear as though the individuals who will receive the greatest weights will be those who do not retake the exam *and* become certified. However, this combination of events cannot occur in our context. Put differently, there are “no defiers” in this environment.

hand side, we have

$$\begin{aligned} \lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds} &= \lim_{s \rightarrow 0^+} \frac{d}{ds} \int [p_c(u)y(1, u) + (1 - p_c(u))y(0, u)] \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u) \\ &= \int [p_c(u)y(1, u) + (1 - p_c(u))y(0, u)] \lim_{s \rightarrow 0^+} \Lambda(s) dF_U(u), \end{aligned} \quad (22)$$

and from the left hand side, we have

$$\begin{aligned} \lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds} &= \lim_{s \rightarrow 0^-} \frac{d}{ds} \int y(0, u) \\ &\quad + p_r(s, u) \Pr(S_2 \geq 0 \mid u, R = 1) p_c(u) (y(1, u) - y(0, u)) \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u) \\ &= \lim_{s \rightarrow 0^-} \int y(0, u) \Lambda(s) dF_U(u) \\ &\quad + \int [y(1, u) - y(0, u)] p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \lim_{s \rightarrow 0^-} \left[\frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} + p_r(s, u) \Lambda(s) \right] dF_U(u). \end{aligned} \quad (23)$$

Then the complete numerator, the difference between (22) and (23), is

$$\begin{aligned} &\lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds} - \lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds} \\ &= \int [y(1, u) - y(0, u)] p_c(u) \times \\ &\quad \left\{ \lim_{s \rightarrow 0^+} \Lambda(s) - \Pr(S_2 \geq 0 \mid u, R = 1) \lim_{s \rightarrow 0^-} \left[\frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} + p_r(s, u) \Lambda(s) \right] \right\} dF_U(u) \\ &= \int [y(1, u) - y(0, u)] \times \left(\lim_{s \rightarrow 0^+} p_c(u) \Lambda(s) \right. \\ &\quad \left. - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \Lambda(s) \right. \\ &\quad \left. - \lim_{s \rightarrow 0^-} \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right) dF_U(u) \end{aligned} \quad (24)$$

$$\begin{aligned} &= \underbrace{\int [y(1, u) - y(0, u)] \cdot \Lambda(0) \cdot \left(\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \right) dF_U(u)}_{(a)} \\ &\quad - \underbrace{\int [y(1, u) - y(0, u)] \left(\lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right) dF_U(u)}_{(b)}. \end{aligned} \quad (25)$$

Consider expanding (a) by plugging in the full expression for $\Lambda(s)$:

$$\int [y(1, u) - y(0, u)] \cdot \frac{f_{S_1|u}(0)}{f_{S_1}(0)} \left[\frac{f'_{S_1|u}(0)}{f_{S_1|u}(0)} - \frac{f'_{S_1}(0)}{f_{S_1}(0)} \right] \left(\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \right) dF_U(u) \quad (26)$$

Under the assumption that s is randomly assigned around the cutoff, there is mean independence between

$$[y(1, u) - y(0, u)] \cdot \left[\frac{f'_{S_1|u}(0)}{f_{S_1|u}(0)} - \frac{f'_{S_1}(0)}{f_{S_1}(0)} \right]$$

and

$$\frac{f_{S_1|u}(0)}{f_{S_1}(0)} \cdot \left(\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \right).$$

Since $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ for mean independent X and Y , (a) is equivalently

$$\int [y(1, u) - y(0, u)] \cdot \left[\frac{f'_{S_1|u}(0)}{f_{S_1|u}(0)} - \frac{f'_{S_1}(0)}{f_{S_1}(0)} \right] dF_U(u) \quad (27)$$

$$\begin{aligned} & \times \int \left(\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \right) \frac{f_{S_1|u}(0)}{f_{S_1}(0)} dF_U(u). \\ & = \int [y(1, u) - y(0, u)] \cdot \left[\frac{f'_{S_1|u}(0)}{f_{S_1|u}(0)} - \frac{f'_{S_1}(0)}{f_{S_1}(0)} \right] dF_U(u) \\ & \times \left(\lim_{s \rightarrow 0^+} \Pr[C = 1 \mid S_1 = s] - \lim_{s \rightarrow 0^-} \Pr[C = 1 \mid S_1 = s] \right), \end{aligned} \quad (28)$$

where the last line follows from equation (17).

With unrestricted heterogeneity in u , it is possible that the first term is positive or negative. However, recall that there is virtually no discontinuity in $\Pr[C = 1 \mid S_1 = s]$ at the cutoff, which is only possible if $\frac{\partial p_r(s, u)}{\partial s} > 0$ for $s < 0$; or in other words, scoring higher on the exam makes one more likely to retake (the “encouragement” effect). Thus, under the assumption there is no discontinuity in $\Pr[C = 1 \mid S_1 = s]$, the entire (a) term is zero.

As a result, the numerator is given by only (b),

$$\int [y(1, u) - y(0, u)] \left(0 - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right) dF_U(u). \quad (29)$$

Denominator. We can similarly take derivatives and limits of the expression for the main component of the RD estimand denominator given by (15) to get the following expression for

the RK estimand denominator:

$$\begin{aligned}
& \lim_{s \rightarrow 0^+} \frac{d\Pr[C = 1 \mid S_1 = s]}{ds} - \lim_{s \rightarrow 0^-} \frac{d\Pr[C = 1 \mid S_1 = s]}{ds} \\
&= \int \Lambda(0) \cdot \left(\lim_{s \rightarrow 0^+} p_c(u) - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) p_r(s, u) \right) \\
&\quad - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} dF_U(u). \tag{30}
\end{aligned}$$

Following similar steps as in the numerator derivation (i.e. plugging in the expression for $\Lambda(s)$ and assuming there is no discontinuity in $\Pr[C = 1 \mid S_1 = s]$ at the cutoff), we can simplify the denominator to

$$\int \left(0 - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right) dF_U(u). \tag{31}$$

Notice that the denominator is negative under the encouragement effect, which is necessary and sufficient for there to be no discontinuity in the likelihood of getting certified—that is, the slope in the likelihood that one becomes certified becomes “flatter” at the cutoff.

Complete expression. Finally we obtain an expression for the RK 2SLS estimand by dividing the numerator (25) by the denominator (31). The resulting expression is again a weighted average of the heterogeneous treatment effects $y(1, u) - y(0, u)$ over the distribution of u ,

$$\frac{\lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds} - \lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[y(C, U) \mid S_1 = s]}{ds}}{\lim_{s \rightarrow 0^+} \frac{d\mathbb{E}[C \mid S_1 = s]}{ds} - \lim_{s \rightarrow 0^-} \frac{d\mathbb{E}[C \mid S_1 = s]}{ds}} = \int [y(1, u) - y(0, u)] \varphi_{RKD}(u) dF_U(u). \tag{32}$$

The weights are given by

$$\varphi_{RKD}(u) = \frac{\left(0 - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right)}{\int \left(0 - \lim_{s \rightarrow 0^-} p_c(u) \Pr(S_2 \geq 0 \mid u, R = 1) \frac{\partial p_r(s, u)}{\partial s} \frac{f_{S_1|u}(s)}{f_{S_1}(s)} \right) dF_U(u)}. \tag{33}$$

We can interpret $\varphi_{RKD}(u)$ as weights because $\varphi_{RKD}(u)$ is non-negative for all u . The RK estimand gives no weight to individuals whose retaking behavior does not respond to their score, thus excluding individuals with such low intrinsic motivation that they would never retake the test, as well as individuals who would always retake the test no matter what. Instead, it gives weight to those who are willing to retake the exam if they were to not pass on their first attempt, and gives relatively more weight to individuals who are more likely to be deterred from retaking if they score far below the cutoff initially (i.e. those who respond more strongly

to the instrument, s).

I.B. Model simulations: From sharp RD to fuzzy RK

This appendix provides a series of simulations that illustrate when and why a kink might arise in place of a discontinuity under dynamic regression discontinuities. First, we show how a RD design approximates a randomized experiment if test-takers were not allowed to retake the test. Second and third, we extend the framework to incorporate the fact that retaking is allowed, first by assuming everyone retakes the exam, and then relaxing this assumption to allow retaking behavior to respond to behavioral incentives. Fourth, we demonstrate that our data is consistent with the model. Fifth, we describe our estimators that give weighted average treatment effects of becoming certified on employment outcomes. We discuss the interpretation of the different weights given by the RD and RKD designs. Sixth, we describe how we implement estimation with the pooled sample of first-time test-takers in our data.

Regression discontinuity under one-shot exams. Our empirical design resembles that of [Card et al. \(2015\)](#) save for two important differences. First, our treatment of interest is binary (getting certified) rather than continuous. Second, in our context, whether one gets treatment is a function of behavioral choices (retaking the exam) rather than being the result of a one-shot evaluation of a deterministic rule.

We begin by describing the one-shot world and add in retaking in the next section. Consider the non-separable model

$$Y = y(C, R, \{S_1, S_2\}, U)$$

where Y denotes the labor market outcome of interest, C is an indicator for getting certified, S_1 and S_2 denotes the two possible scores on the Praxis on various attempts where S_2 is only observed if the individual chooses to retake (i.e. $R = 1$), and U captures individual heterogeneity. Individuals have some underlying test-taking ability μ_i such that their scores are determined by $S_{ai} = \mu_i + \varepsilon_i$ for all attempts a and mean zero ε_i . In other words, they do not have precise control over their score.

The treatment of interest is whether one gets certified: $C = C(\{S_1, S_2\}, R, \xi)$,⁶⁶ which can

⁶⁶The analogous expression in [Card et al. \(2015\)](#) is the “policy function” or “treatment assignment rule” underlying the fuzzy regression kink design. However, in the case of Card et al., the policy function is a deterministic rule that exhibits a kink at some point in the support of the underlying variable that determines C , whereas in our setting the function contains a random component.

only happen if one passes on their first attempt ($S_1 \geq 0$) or on their second attempt ($R = 1$ and $S_2 \geq 0$). Even if one passes, there is additional randomness ξ_i that can influence certification completion, which may be correlated with U_i and thus also with Y_i .

Suppose for now that retaking is not allowed. The top row of Figure II shows simulations of what the first stage relationships between first-attempt scores (S_1) and the likelihood of passing the test or becoming certified would look like.⁶⁷ Those who do not pass can never pass the exam nor become certified, generating clean discontinuities in the likelihood of passing and of getting certified at $S_1 = 0$.

We are interested in the effect of getting certified on earnings. Regressing earnings on whether one gets certified would give a biased estimate of the parameter of interest, since the decision to get certified involves choices that may be made using information that is endogenous to earnings (i.e. $\mathbb{E}[\xi'U] \neq 0$).

However, notice that in this world, we can use whether one passes the Praxis on their first attempt, $\mathbb{I}(S_1 \geq 0)$, as a valid instrument for whether one gets certified. First, because the test can only be attempted once, passing the Praxis strongly affects the likelihood of certification. Secondly, the exclusion restriction holds: since individuals around the threshold are exogenously assigned permission to continue down the teacher pipeline and the Praxis is only relevant for teaching accreditation, the only channel through which we would expect passing the Praxis to affect earnings is through completing the teacher pipeline.

Dynamic regression discontinuities under automatic retaking. We now extend the one-shot framework to allow for the real-world possibility of retaking. For exposition, assume that individuals have are endowed with at most two retakes and that there is no cost to retaking the exam, such that all individuals who fail on their first try retake ($R = 1$) and realize their S_2 .

The middle row of Figure II shows the resulting simulated first-stages under retaking. As in the one-shot case, those who pass on their first attempt have a constant probability of becoming certified. Furthermore, under the data-generating process $S_{ai} = \mu_i + \varepsilon_i$, those with higher first-attempt scores are more likely to have high μ_i . Thus, for retakers, the probability that they will pass on their second attempt and become certified is increasing in their initial

⁶⁷We assume that among those who pass, high-scorers are no more likely than barely-passers to enroll in and complete an EPP. However, as we know from the descriptives in Section ??, many who pass the Praxis do not continue through an EPP. We therefore choose parameters in our simulation to replicate this observation, with passers having a 2/3 probability of becoming certified.

score S_{1i} . The result visually is a discontinuity and now also a kink in the likelihood of passing and of getting certified at the cutoff.

Since the discontinuity is still present and pronounced, similarly to the one-shot world we can still use whether one passes the Praxis on their first attempt, $\mathbb{I}(S_1 \geq 0)$, as a valid instrument for whether one gets certified.

Dynamic regression discontinuities under behavioral retaking. Finally, we extend the retaking framework once more by relaxing the assumption that individuals who fail automatically retake. We now allow for the possibility that not everyone who fails the exam initially will retake.

Assume that for those who fail initially, retaking behavior is influenced by the “encouragement” effect: Getting a score closer to the cutoff increases the likelihood one retakes the exam. Under this assumption, individuals with low intrinsic motivation to get certified and/or individuals who are prone to being discouraged by getting a realized score below the cutoff will not retake the exam. Those who retake will be highly intrinsically motivated and/or encouraged by having scored close to the cutoff on their first attempt.

The last row of Figure II shows the resulting simulated first-stages under the two behavioral assumptions above. As before, those with higher first-attempt scores are more likely to have high μ_i . But in addition, because retakers are selected on their motivation to get certified and are influenced by the random assignment of their initial score, the slope in the likelihood of certification is steeper to the left of the cutoff. The result visually a stronger kink in the likelihood of passing and of getting certified at the cutoff. In particular, there is a diminished discontinuity in the likelihood of getting certified at the cutoff, which could be zero or even negative depending on the strength of the “intrinsic motivation” effect.

In this environment, the diminished or lack of discontinuity renders the instrument $\mathbb{I}(S_1 \geq 0)$ weak or invalid. However, we can construct a valid instrument that leverages both (1) the fact that scores are randomly assigned near the cutoff *and* (2) the actual value of the score influences retaking behavior: $\mathbb{I}(S_1 \geq 0) \times s$. As is evident by the kink, the instrument is relevant: near the cutoff, those who pass are randomly assigned a lower cost of continuing on to certification, while those who fail are randomly assigned a higher cost that increases with the distance of S_1 from the cutoff. The exclusion restriction would be threatened if the act of retaking the exam influences earnings through a channel other than certification, such as by improving one’s human capital through the studying process alone. However, given that

the exams test general core skills, it is unlikely that this channel exists. Thus, we assume the exclusion restriction holds as well.

Reconciling the model and data. The last row of Figure II can be contrasted with the actual first-stages in the data shown in Figure I. The actual first-stages in our data are consistent with the model, especially in the kinked shape of the functions. In addition, the discontinuity in the likelihood of certification in the data is virtually zero, suggesting that retaking behavior is affected by both the “encouragement” effect and the “intrinsic motivation” effect.

I.C. Re-casting the LATE framework with retaking

Because we estimate our RK estimand using a 2SLS estimator, it must have a LATE interpretation (Angrist et al.). In this section, we define what it means to be a “complier” when there can be multiple attempts at becoming eligible for treatment.

Typically, when treatment is assigned based on a one-shot deterministic rule, the union of actions taken by always-takers, never-takers, and compliers account for the full set of possible actions. If individuals could only attempt the Praxis once, then compliers would be individuals who become certified if they pass on their first attempt but do not become certified if they fail on their first attempt. Always-takers in this setting are individuals who, regardless of whether they pass or fail on the first attempt, will find a way to obtain certification. It is only possible to obtain certification by retaking the exam. Thus, always-takers are individuals who either pass on their first attempt or would retake the exam until they pass. Never-takers in our setting are individuals who will never become certified, regardless of whether they pass or fail on their first attempt.

However, when individuals have multiple chances to influence their eligibility to obtain treatment, the three classic types of behavior of always-takers, never-takers, and compliers no longer cover all possible actions. In fact, in settings like ours, individuals can take actions that do not fit any of the typical definitions of always-takers, never-takers, and compliers. For example, consider individuals who pass on their first attempt, retake once, but do not get certified. These individuals could be thought of as never-takers, but they could also be thought of as individuals who are unwilling to take the test more than once, or who are discouraged by their score on the second exam and therefore do not continue. There is no natural definition for such behavior in the classic LATE framework.

We therefore re-cast the concepts of compliers, always-takers, and never-takers with respect to retaking behavior. We define a “retaking complier” as someone whose retaking behavior is responsive to the first-attempt score they receive. Compliers do not retake if they pass on their first attempt, but otherwise retake with positive probability that is increasing in their first-attempt score. We define a “retaking always-taker” as someone who always retakes if they fail on their first attempt. We define a “retaking never-taker” as someone who never retakes if they fail on their first attempt. Finally, we define a “retaking defier” as someone who only retakes if they pass on their first attempt. We (sensibly) do not observe defiers in our data.

The identification proofs in Supplementary Materials [I.A](#) show that the RK estimator we use only gives positive weight to “retaking compliers.” The data suggests that the majority of test-takers are compliers: the retaking rate among those who fail on the first attempt is high (80%) and there is a significant positive relationship between the likelihood of retaking and initial scores.

I.D. Additional Survey Details

Implementation. The survey was coded using the online survey tool Qualtrics, which allows users to send unique links to the survey out to respondents via email. To target KY teachers, we obtained a complete list of school staff and their email addresses through MCH Strategic Data and emailed unique Qualtrics survey links directly to staffs’ school email addresses.

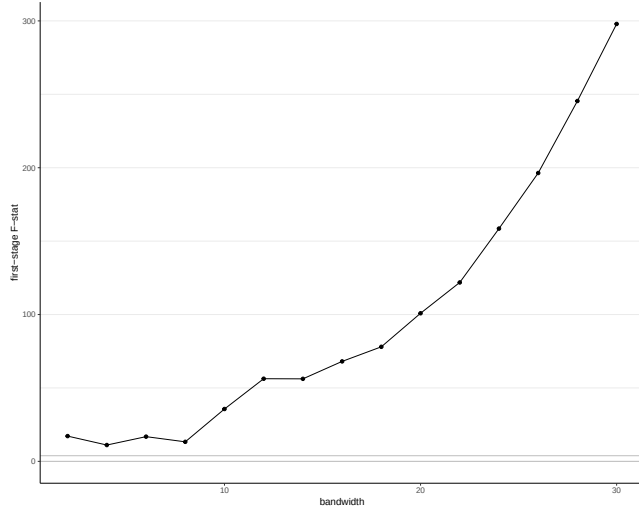
Trick questions. To identify inattentive respondents, we include two attention check questions at the start of the first and second third of the survey, one of which occurs in between the first and second stated-preference experiment. The first question uses the “reverse wording” technique: it asks a question that was already asked in the survey, but changes the direction of the multiple choice scale. Those who provide inconsistent responses to the true question and the reverse wording question are labelled as “inattentive.” The second question is similar to one that was used in the AWCS: the text contains a plausibly genuine question about job preferences but also specifies in the introduction of the text that the respondent should respond in a specific way, regardless of their true answer to the question. Those who do not respond in the specified manner are labelled as “inattentive.”

Presenting the wages. To facilitate ease of interpretation for the respondent, we display the earnings in the hypothetical job choice in terms of the units in which the respondent reported

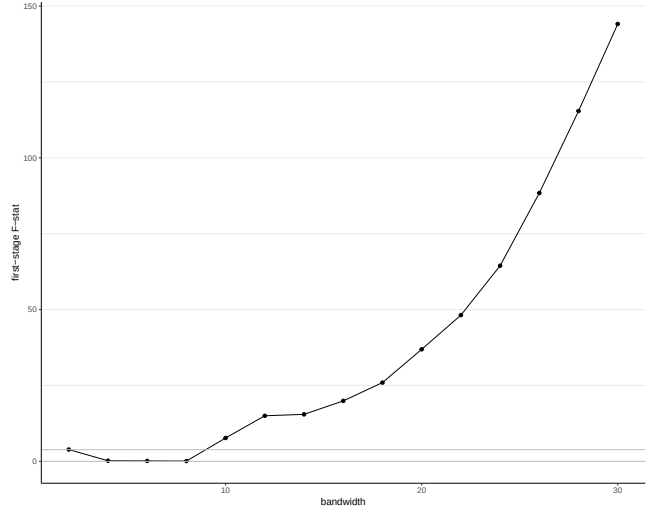
their earnings (hourly, weekly, biweekly, monthly, or annually), and round to the nearest \$.50 if hourly, \$10 if weekly, biweekly, or monthly, and \$100 if annually. To convert hourly wages to annual earnings, we assumed the job constituted 52 weeks of work. If the weekly work hours exceeded 40, we called the excess time overtime pay and assigned it a wage of 1.5 times the randomly assigned hourly wage. If presenting hourly wages, we also presented the implied weekly earnings below, including overtime pay.

Limiting cases where one jobs dominates another. We used the willingness-to-pay estimates from the college-educated workers in [Maestas et al. \(2023\)](#) as a benchmark with which to rank job attribute values, but not the attributes themselves. When one of the randomly generated jobs dominated the other on all aspects, we redrew the scaling parameters θ_A and θ_B and recomputed the wage offered. If one job still dominated, we redrew the attribute values. These steps reduced the likelihood that one job would dominate the other in all aspects.

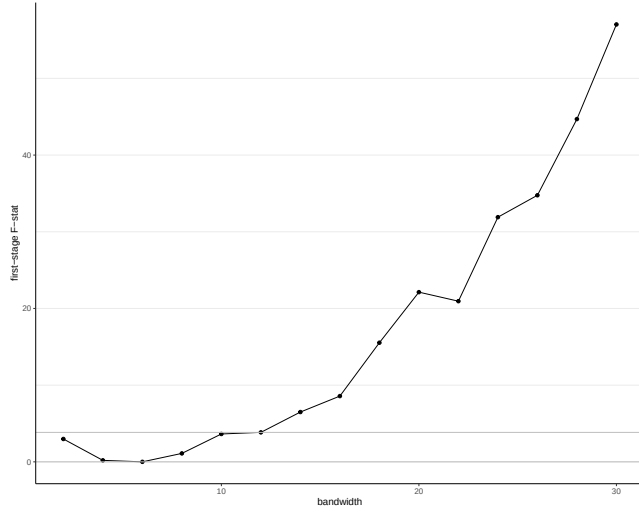
I.E. Additional Figures



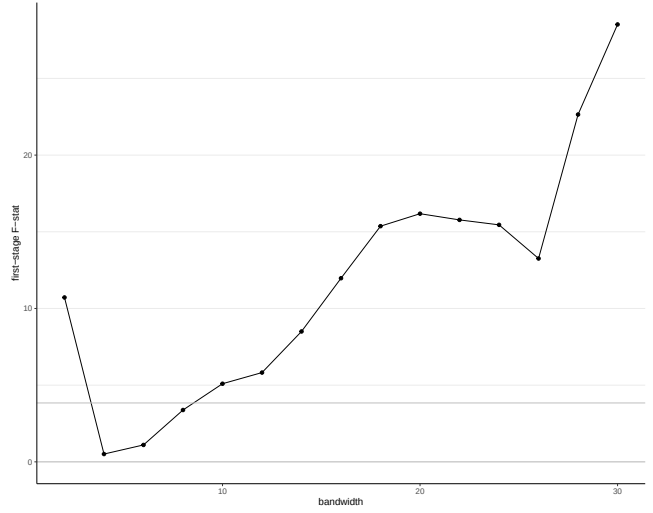
(a) Ever-passing the math test.



(b) Ever-passing all 3 tests.

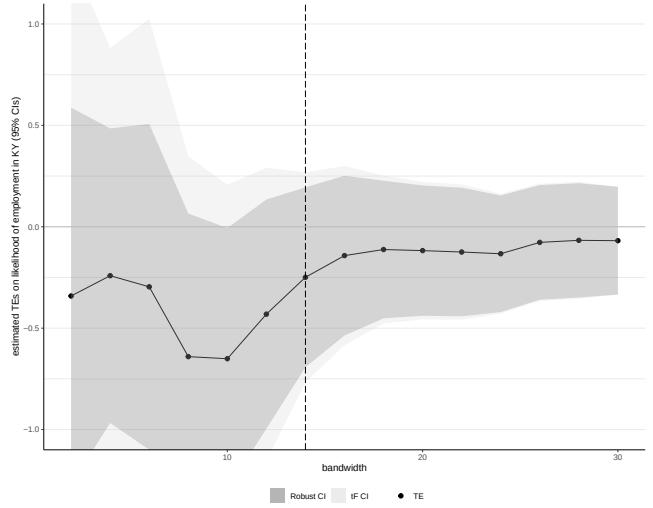
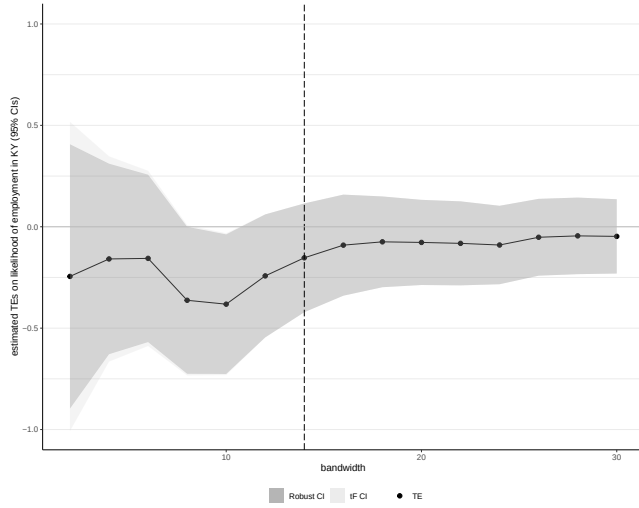


(c) Becoming certified within 4yrs.



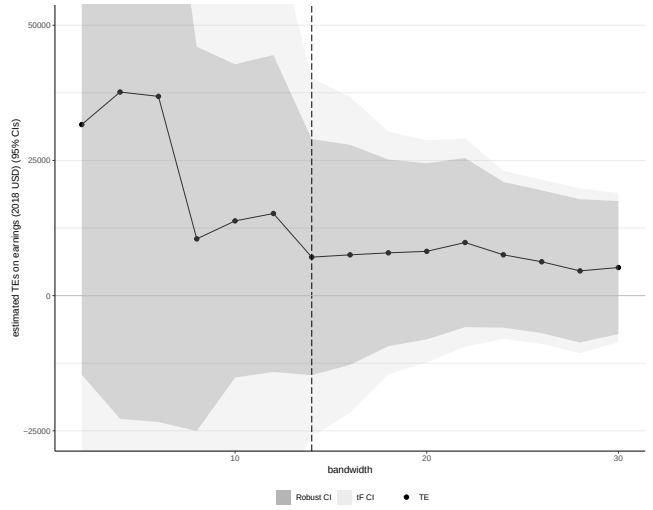
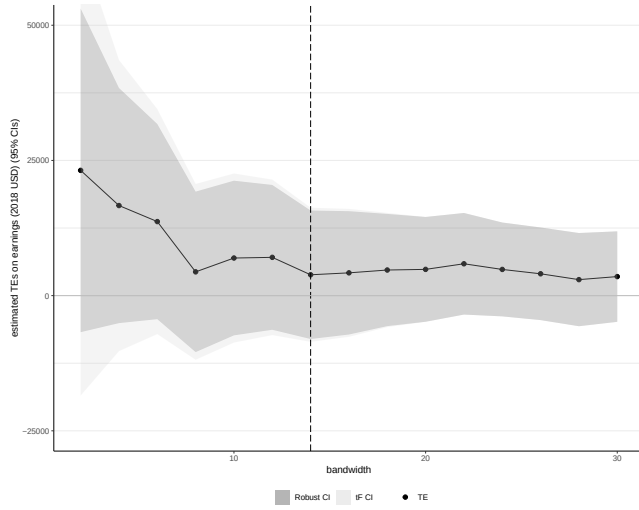
(d) Being a teacher in 4yrs.

Figure A.1: Sensitivity of first-stage RKD F-stats to bandwidth selection, for each stage along the teacher pipeline.



(a) Effect of ever-passing math on $\Pr(\text{employed})$. (b) Effect of ever-passing all 3 on $\Pr(\text{employed})$.

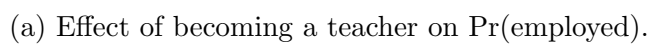
Figure A.2: RD estimates of the effect on the likelihood of being employed (in the state of KY) of passing the Praxis, over bandwidths ranging from 2-30.



(a) Effect of ever-passing math on earnings.

(b) Effect of ever-passing all 3 on earnings.

Figure A.3: RD estimates of the effects on earnings in 4yrs of passing the Praxis, over bandwidths ranging from 2-30.



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* Imagine you are offered the two jobs shown below. (1 of 3)

The differences between the jobs are **highlighted in yellow**. The jobs are **otherwise exactly the same**, even on characteristics not listed in the table.

Please review the jobs and indicate below whether you prefer Job A or Job B.

	Job A	Job B
Hours	Full-Time - 40 hours per week	Full-Time - 40 hours per week
Control Over Hours	Set your own schedule	Schedule set by manager
Option to Telecommute	Yes	Yes
Physical Demands	Heavy physical activity	Heavy physical activity
Pace	Relaxed	Relaxed
Independence	You can choose how you do your work	You can choose how you do your work
Paid Time Off (Vacation and Sick Days)	20 days	10 days
Working With Others	Team-based and evaluated on performance of team	Team-based and evaluated on performance of team
Training	You already have the skills for this job	You already have the skills for this job
Impact on Society	Occasional opportunities to make a positive impact on your community or society	Occasional opportunities to make a positive impact on your community or society
Pay	\$56,500 per year	\$62,800 per year

	Strongly Prefer Job A	Prefer Job A	Prefer Job B	Strongly Prefer Job B
Which job do you prefer?	☐	☐	☐	☐

Figure A.5: Screenshot of hypothetical job pair from survey to Kentucky teachers

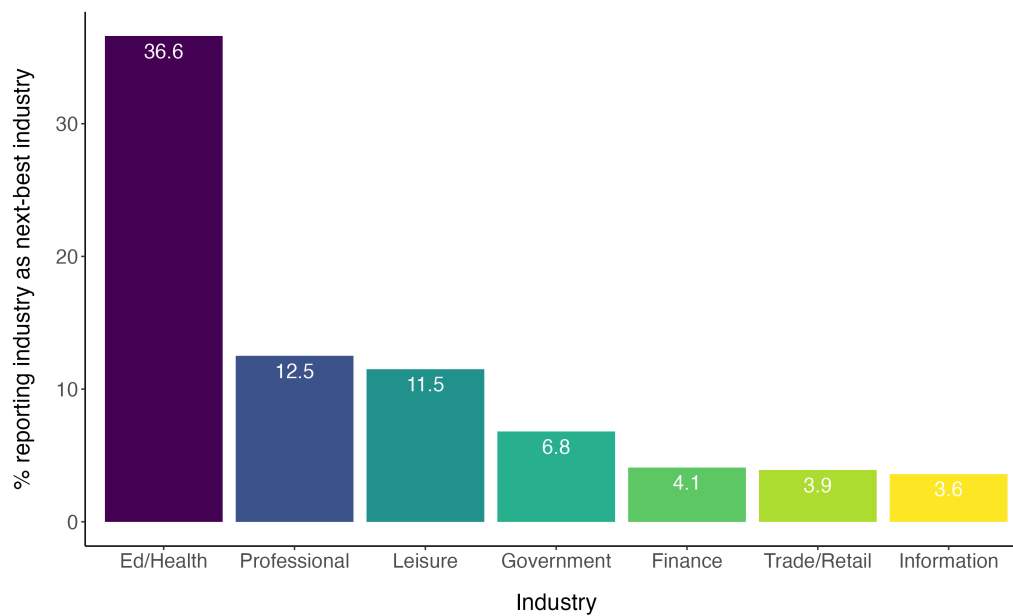


Figure A.6: Most Popular Next-Best Industries Reported by Kentucky Teachers

Note: This figure shows the distribution of responses to the following survey question: “If you were not a teacher today, what alternative job might you be doing instead? Please visualize a job that you think is realistic, such as a job you may have applied to before, or a job you or your colleagues may have worked in besides teaching. Which industry would this job be in?”

I.F. Additional Tables

Table A.1: Incidence of Working Conditions in Teaching and the AWCS

	KY Teachers (1)	AWCS		
		All workers (2)	Women (3)	College educ. (4)
Contractual hours per week	38.8 (4.5)	39.6 (9.9)	37.3 (9.6)	40.2 (10.2)
Total hours per week	49.9 (9.8)			
Hourly wage (in 2015 USD)	28.4 (11.9)	30.3 (36.6)	26.0 (28.0)	38.0 (30.1)
<i>% with each working condition</i>				
Sets own schedule	10.3	56.5	58.8	67.5
Telecommute	4.1	36.4	37.0	54.9
Moderate physical activity	87.6	38.4	38.1	36.7
Mostly sitting	2.5	42.9	49.4	57.9
Relaxed pace	12.8	29.8	32.6	35.5
Choose how do work	81.1	86.4	86.4	91.8
1-14 paid time off (PTO) days	94.3	26.0	25.1	21.3
15+ PTO days	4.1	59.7	61.2	68.7
Team-based, evaluated on own	55.9	49.1	51.3	52.1
Work by self	41.1	32.4	35.1	33.5
Training opportunities	54.0	70.0	65.4	74.0
Frequent opp. positive impact	61.3	34.5	39.5	39.0
Observations	554	1738	968	908

Note: This table presents summary statistics on the jobs held by survey respondents in my survey (Column 1) and in the AWCS (Columns 2–4). The only variable that does not have a direct comparison across columns is “total hours per week,” which was not asked in the AWCS. Standard deviations are shown in parentheses.

Table A.2: Comparison of Working Conditions in Teaching to Jobs in Other Industries

	Teacher (1)	Industry in AWCS						
		Ed/Health (2)	Prof (3)	Leisure (4)	Govt (5)	Finance (6)	Trade (7)	Info (8)
Next-best industry %	-	36.6	12.5	11.5	6.8	4.1	3.9	3.6
Mean hours per week	39.0	38.3	39.7	31.8	40.0	40.0	40.6	41.9
Mean hourly wage (in 2015)	25.9	30.8	34.1	16.3	32.5	37.0	33.6	29.6
<i>% with each working condition</i>								
Sets own schedule	14.7	56.3	64.8	66.5	55.8	76.0	51.7	70.6
Telecommute	13.4	32.6	52.8	21.9	38.9	58.4	21.1	61.9
Moderate physical demands	88.2	50.1	32.3	54.4	38.4	14.9	39.0	21.7
Mostly sitting	5.8	36.3	50.6	13.2	54.1	81.5	33.2	77.4
Relaxed pace	13.9	33.5	33.2	13.8	30.7	27.1	23.4	25.3
Choose how do work	93.5	90.4	88.0	86.0	93.4	86.7	75.9	96.3
1-14 PTO days	61.2	26.8	30.4	21.9	19.7	17.9	38.8	12.3
15+ PTO days	28.6	61.1	59.0	28.0	75.2	74.5	49.9	68.1
Team-based, evaluate own	68.1	56.3	43.6	58.1	50.7	43.0	45.8	55.8
Work by self	26.0	27.5	38.8	25.0	37.4	37.6	31.0	31.7
Training opportunities	63.3	67.3	73.9	57.9	78.8	70.5	66.8	70.8
Frequent opp. positive impact	61.8	51.9	25.6	31.8	49.0	35.9	26.3	19.2
N	46	385	241	59	152	146	233	52

Note: This table presents comparisons of the jobs held by Kentucky teachers in my survey (Column 1) and various industries in the AWCS in 2015 (Columns 2-8), similarly to Table A.1. The first row includes an additional statistic: the share of Kentucky teachers surveyed who reported that their next-best job would be in the column's industry.

Table A.3: Comparison of Working Conditions in Teaching to Below-Average Paying Jobs in Other Industries in 2015 (AWCS)

	Teacher (1)	Industry in AWCS						
		Ed/Health (2)	Prof (3)	Leisure (4)	Govt (5)	Finance (6)	Trade (7)	Info (8)
Next-best industry %	-	35.9	12.5	11.5	6.8	4.1	3.9	3.6
Mean hours per week	39.0	37.9	39.5	33.8	38.2	39.0	40.2	36.9
Mean hourly wage (in 2015)	25.9	18.1	18.8	10.2	21.8	20.2	17.1	18.4
<i>% with each working condition</i>								
Sets own schedule	14.7	48.1	54.7	69.0	52.4	71.1	45.2	67.9
Telecommute	13.4	23.3	37.5	12.7	29.9	46.3	16.1	52.3
Moderate physical demands	88.2	56.6	32.3	54.4	40.6	19.5	42.6	37.6
Mostly sitting	5.8	28.8	43.7	3.9	52.8	75.7	26.8	60.4
Relaxed pace	13.9	37.4	33.8	5.5	30.7	31.8	22.3	18.3
Choose how do work	93.5	91.6	83.6	87.0	92.8	88.8	77.5	97.2
1-14 PTO days	61.2	31.0	37.4	25.5	26.8	21.1	44.4	8.9
15+ PTO days	28.6	54.2	50.4	20.5	65.8	69.9	43.1	62.4
Team-based, evaluate own	68.1	52.1	40.6	63.4	54.0	40.4	45.0	54.3
Work by self	26.0	27.8	41.1	25.0	38.8	41.7	30.3	43.2
Training opportunities	63.3	67.0	67.6	50.4	73.8	65.3	68.7	71.3
Frequent opp. positive impact	61.8	51.2	20.8	29.4	51.1	38.6	24.1	21.9
N	46	271	169	30	96	100	188	25

Table A.4: Working Conditions in the U.S. in 2015, for Teachers and Comparison Demographic Groups

	Overall (1)	Teachers (2)	Women (3)	College-Educ. (4)
Mean hours per week	39.62 (9.89)	39.03 (10.09)	37.32 (9.58)	40.23 (10.17)
Mean wage (in 2015 \$)	30.30 (36.55)	25.89 (15.02)	25.99 (27.95)	38.00 (30.10)
<i>% with each working condition</i>				
Sets own schedule	56.5	14.7	58.8	67.5
Telecommute	36.4	13.4	37.0	54.9
Moderate physical activity	38.4	88.2	38.1	36.7
Mostly sitting	42.9	5.8	49.4	57.9
Relaxed pace	29.8	13.9	32.6	35.5
Choose how do work	86.4	93.5	86.4	91.8
1-14 Paid Time Off (PTO) days	26.0	61.2	25.1	21.3
15+ PTO days	59.7	28.6	61.2	68.7
Team-based, evaluated on own	49.1	68.1	51.3	52.1
Work by self	32.4	26.0	35.1	33.5
Training opportunities	70.0	63.3	65.4	74.0
Frequent opp. to serve	34.5	61.8	39.5	39.0
Observations	1738	46	968	908

Table A.5: Regressions of Working Conditions on Demographic Variables (part 1)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sets own sched.	Telecommute	Heavy Physical Activity	Moderate P.A.	Mostly sitting	Relaxed pace	Choose how do work
Teacher	-58.25 (8.44)	-42.98 (8.02)	1.17 (6.49)	55.92 (8.53)	-57.09 (8.40)	-21.62 (8.02)	1.46 (6.03)
Female	3.30 (2.49)	-1.08 (2.36)	-8.01 (1.91)	-3.75 (2.51)	11.76 (2.47)	5.56 (2.36)	-0.50 (1.78)
Nonwhite	-7.66 (3.17)	-6.86 (3.01)	0.57 (2.44)	2.85 (3.21)	-3.42 (3.16)	-0.18 (3.02)	-2.47 (2.27)
HS or less	-28.35 (3.00)	-39.48 (2.85)	24.88 (2.30)	6.35 (3.03)	-31.23 (2.98)	-12.69 (2.85)	-12.53 (2.14)
Some college	-17.05 (3.07)	-30.37 (2.92)	13.55 (2.36)	8.42 (3.10)	-21.97 (3.06)	-5.97 (2.92)	-4.17 (2.19)
Under 35	-2.64 (4.78)	-6.86 (4.54)	2.66 (3.68)	-4.83 (4.83)	2.17 (4.76)	-21.58 (4.55)	12.82 (3.42)
Aged 35-49	-4.82 (4.53)	-3.04 (4.31)	3.76 (3.49)	-14.20 (4.58)	10.44 (4.51)	-11.46 (4.31)	5.47 (3.24)
Aged 50-61	-3.39 (4.63)	-3.26 (4.40)	1.53 (3.56)	-8.37 (4.68)	6.84 (4.61)	-9.28 (4.40)	8.36 (3.31)
N	1,492	1,492	1,492	1,492	1,492	1,492	1,492

Table A.6: Regressions of Working Conditions on Demographic Variables (part 2)

	(1) No PTO	(2) 1-14 PTO Days	(3) 15+ PTO days	(4) Team Eval.	(5) Team-based, Own Eval.	(6) Work by self	(7) Training opps.	(8) Freq. opp. to serve
Teacher	0.16 (5.94)	41.94 (7.82)	-42.11 (8.58)	-7.22 (6.93)	16.27 (8.89)	-9.05 (8.35)	-10.05 (7.94)	21.56 (8.38)
Female	-0.79 (1.75)	-1.36 (2.30)	2.14 (2.53)	-9.17 (2.04)	2.89 (2.62)	6.28 (2.46)	-11.39 (2.34)	9.39 (2.47)
Nonwhite	-4.31 (2.23)	6.12 (2.94)	-1.81 (3.22)	3.95 (2.60)	0.49 (3.34)	-4.44 (3.14)	8.90 (2.98)	-2.66 (3.15)
HS or less	6.07 (2.11)	15.00 (2.77)	-21.07 (3.04)	5.73 (2.46)	-5.28 (3.15)	-0.45 (2.96)	-12.54 (2.82)	-9.74 (2.97)
Some college	3.33 (2.16)	9.38 (2.84)	-12.70 (3.12)	7.09 (2.52)	-4.33 (3.23)	-2.75 (3.04)	-6.16 (2.89)	-4.85 (3.05)
Under 35	-6.91 (3.36)	7.20 (4.43)	-0.29 (4.86)	3.65 (3.93)	4.69 (5.03)	-8.34 (4.73)	19.57 (4.50)	-1.65 (4.75)
Aged 35-49	-9.59 (3.19)	3.60 (4.20)	5.99 (4.61)	7.83 (3.72)	-5.89 (4.77)	-1.93 (4.48)	10.72 (4.27)	5.24 (4.50)
Aged 50-61	-8.18 (3.26)	-1.47 (4.29)	9.65 (4.70)	3.95 (3.80)	-4.55 (4.87)	0.59 (4.58)	7.34 (4.36)	1.63 (4.60)
N	1,492	1,492	1,492	1,492	1,492	1,492	1,492	1,492

Table A.7: Willingness-to-pay estimates for job attributes

	Teachers (1)	AWCS		
		All workers (2)	College-educ. (3)	Ed/Health, < median wage (4)
Sets own schedule <i>vs. Schedule set by manager</i>	0.080 (0.018)	0.089 (0.007)	0.104 (0.008)	0.071 (0.017)
Telecommute <i>vs. No telecommuting</i>	0.063 (0.015)	0.042 (0.007)	0.069 (0.008)	0.050 (0.018)
Moderate physical activity <i>vs. Heavy physical activity</i>	0.159 (0.021)	0.145 (0.010)	0.168 (0.013)	0.197 (0.026)
Sitting <i>vs. Heavy physical activity</i>	0.119 (0.026)	0.116 (0.010)	0.137 (0.013)	0.142 (0.030)
Relaxed <i>vs. Fast pace</i>	0.093 (0.017)	0.043 (0.007)	0.052 (0.007)	0.042 (0.019)
Choose how do work <i>vs. Tasks well defined</i>	0.050 (0.017)	0.040 (0.007)	0.059 (0.007)	0.050 (0.019)
10 days PTO <i>vs. No days PTO</i>	0.196 (0.024)	0.164 (0.009)	0.158 (0.010)	0.197 (0.025)
20 days PTO <i>vs. No days PTO</i>	0.268 (0.025)	0.230 (0.010)	0.226 (0.011)	0.289 (0.031)
Team-based, evaluate own <i>vs. Team-based, evaluate team</i>	0.099 (0.021)	0.065 (0.010)	0.074 (0.011)	0.067 (0.027)
Work by self <i>vs. Team-based, evaluate team</i>	0.048 (0.022)	0.086 (0.010)	0.073 (0.012)	0.097 (0.026)
Training opportunities <i>vs. Already have skills</i>	0.041 (0.016)	0.054 (0.007)	0.064 (0.008)	0.044 (0.017)
Frequent opp. to serve <i>vs. Occasional opp. to serve</i>	0.043 (0.015)	0.036 (0.007)	0.039 (0.008)	0.050 (0.020)
N	1,662	17,380	9,080	2,710

Note: Column 1 shows estimates of Kentucky teachers' willingness to pay using data from the stated preference experiments. Column 2 shows the same estimates for all workers in the AWCS; Column 3 shows the estimates for college educated workers in the AWCS; and Column 4 shows the estimates for workers in the education or health sectors in the AWCS earning below the median wage in the two sectors. Standard errors, shown in parentheses, are estimated using the delta method and are clustered at the respondent level.

Table A.8: OLS estimates of cross-sectional differences in earnings between teachers and non-teachers among KY college graduates

	<i>Dependent variable: Annual total wages (2018 USD)</i>					
	Sample: 18-60yo		Sample: 18-60yo, working at least 4qtrs		Sample 18-30yo, working at least 4qtrs	
	(1)	(2)	(3)	(4)	(5)	(6)
Is teacher	4,977.318*** (71.502)	-6,187.064*** (61.972)	-1,649.273*** (69.929)	-7,252.242*** (66.209)	3,218.845*** (90.019)	2,545.662*** (83.769)
Female		-7,643.454*** (43.785)		-8,566.723*** (50.745)		-4,744.887*** (58.738)
Black		-4,958.967*** (141.189)		-6,057.636*** (168.967)		-3,967.194*** (201.492)
White		463.919*** (114.653)		19.382 (138.385)		1,204.935*** (162.909)
Hispanic		-2,796.032*** (132.946)		-3,140.671*** (155.009)		-1,512.377*** (180.965)
Age (years)		589.780*** (2.885)		654.235*** (3.307)		2,337.656*** (16.385)
Nr quarters employed		14,685.510*** (26.678)				
Holds at least a Master's		14,090.470*** (49.858)		15,369.000*** (56.726)		10,167.130*** (83.056)
Mean	46591.01					
Observations	1,201,888	1,201,888	991,951	991,951	387,848	387,848
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.004	0.355	0.001	0.218	0.003	0.175

Note: Sample includes people 18-60yo with at least a BA earned in KY since 2005. *p<0.1; **p<0.05; ***p<0.01